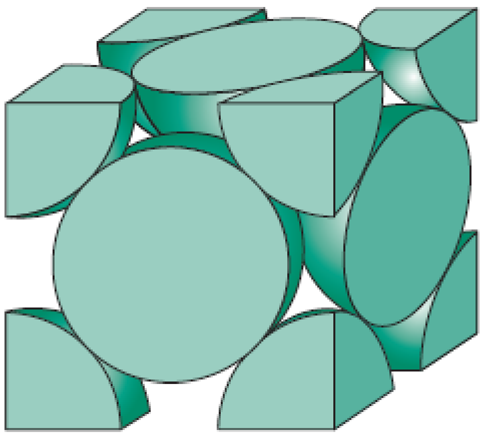
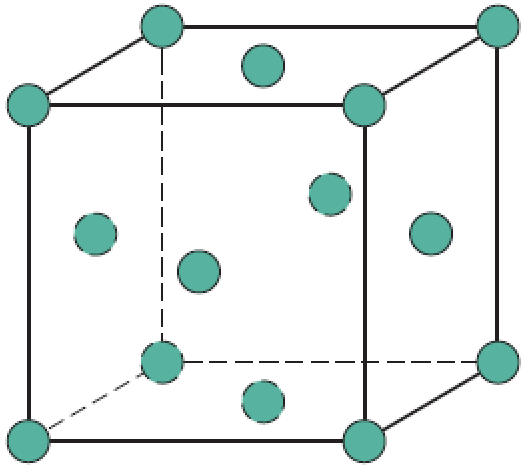
Crystal Structures - Fundamental concept

- A **crystalline material** is one in which the *atoms are situated in a repeating or periodic array over large atomic distances*; that is, long-range order exists, such that upon solidification.
- Periodicity is one of the most important properties of crystals.
- Crystals are *highly symmetrical arrays of atoms* which substantially simplifies their analysis
- Some of the properties of crystalline solids depend on the **crystal structure** of the material, the manner in which atoms, ions, or molecules are spatially arranged.



(a)



(b)

Crystal Lattice

- Crystal lattice is the *periodic and systematic arrangement of atoms* that are found in crystals with the exception of amorphous solids and gases.

- In the simplest of terms, the *crystal lattice can be considered as the points of intersection between straight lines in a three-dimensional network*.

- The physical properties of crystals like cleavage, electronic band structure and optical transparency are predominantly governed by the crystal lattice.

- *Lattice – Arrangement of atoms/ions*

- *Motif/Basis – A group of one /more atoms located in a particular way with respect to each other and associated with lattice point*

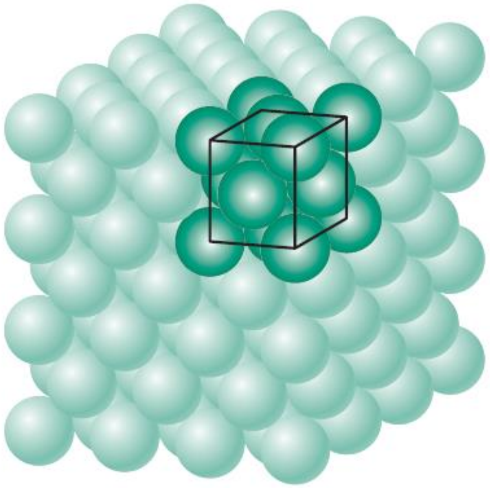
- *Crystal Structure = Lattice + Motif/Basis*

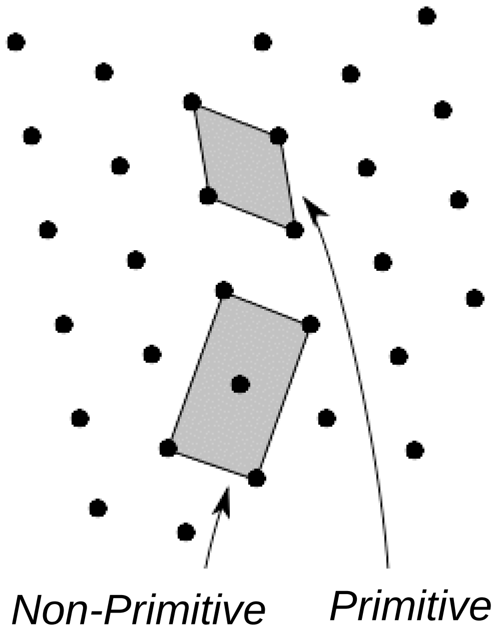
Crystal Lattices

- The concept of a ***lattice*** is directly related to the idea of translational symmetry. ***A lattice is a network or array composed of single motif which has been translated and repeated at fixed intervals throughout space***. For example, a square which is translated and repeated many times across the plane will produce a planar square lattice.
- The ***unit cell*** of a lattice is *the smallest unit which can be repeated in three dimensions* in order to construct the lattice. In a crystal, the unit cell consists of a specific group of atoms which are bonded to one another in a set geometrical arrangement. This unit and its constituent atoms are then repeated over and over in order to construct the crystal lattice. The surroundings in any given direction of one corner of a unit cell must be identical to the surroundings in the same direction of all the other corners.
- *The corners of the unit cell therefore serve as points which are repeated to form a lattice array; these points are termed* ***lattice points***.
- The vectors which connect a straight line of equivalent lattice points and delineate the edges of the unit cell are known as the ***crystallographic axes***.

Unit Cell

- represent the symmetry of the crystal structure
- the *basic structural unit or building block of the crystal structure* and defines the crystal structure by virtue of its geometry and the atom positions within.
- The unit cell is characterized by its lattice parameters which consist of the length of the cell edges and the angles between them
- Unit cell is the subdivision of the lattice that still retains the overall characteristics of the entire lattice
- By stacking identical unit cells, the entire lattice can be constructed
- Primitive unit cells contain only one lattice point, which is made up from the lattice points at each of the corners.
- Non-primitive unit cells contain additional lattice points, either on a face of the unit cell or within the unit cell, and so have more than one lattice point per unit cell.





Symmetry restricts the unit cells to certain shapes so that the entire space is covered without gaps and overlaps

Crystal Symmetries

- Crystals possess a *regular, repetitive internal structure*.
- The concept of symmetry describes the repetition of structural features. Crystals therefore possess symmetry, and much of the discipline of crystallography is concerned with describing and cataloging different types of symmetry.
- Two general types of symmetry exist.
  - ***Translational symmetry*** describes the periodic repetition of a structural feature across a length or through an area or volume.
  - ***Point symmetry*,** on the other hand, describes the periodic repetition of a structural feature around a point. Reflection, rotation, and inversion are all point symmetries.

- Symmetries are most frequently used to classify the different crystal structures.
- In general one can generate 14 basic crystal structures through symmetries. These are called ***Bravais lattices*.** Any crystal structures can be reduced to one of these *14 Bravias lattices*.

Ideal Crystals → Real Crystals → Microstructures → Material → Component

Ideal Crystal — Ideal Mathematical Crystal

↓ Consider only the Geometrical Entity or only the Physical Property

Crystal*

↓ Consider only the Orientational or Positional Order

Crystal**

↓ Put in Crystalline defects & Free Surface & Thermal Vibration

'Real Crystal'

↓ Put in Multiple Crystals **(Phases)** giving rise to interfacial defects

~Microconstituents

↓ Put multiple ~microconstituents Add additional residual stress†

Microstructure

↓

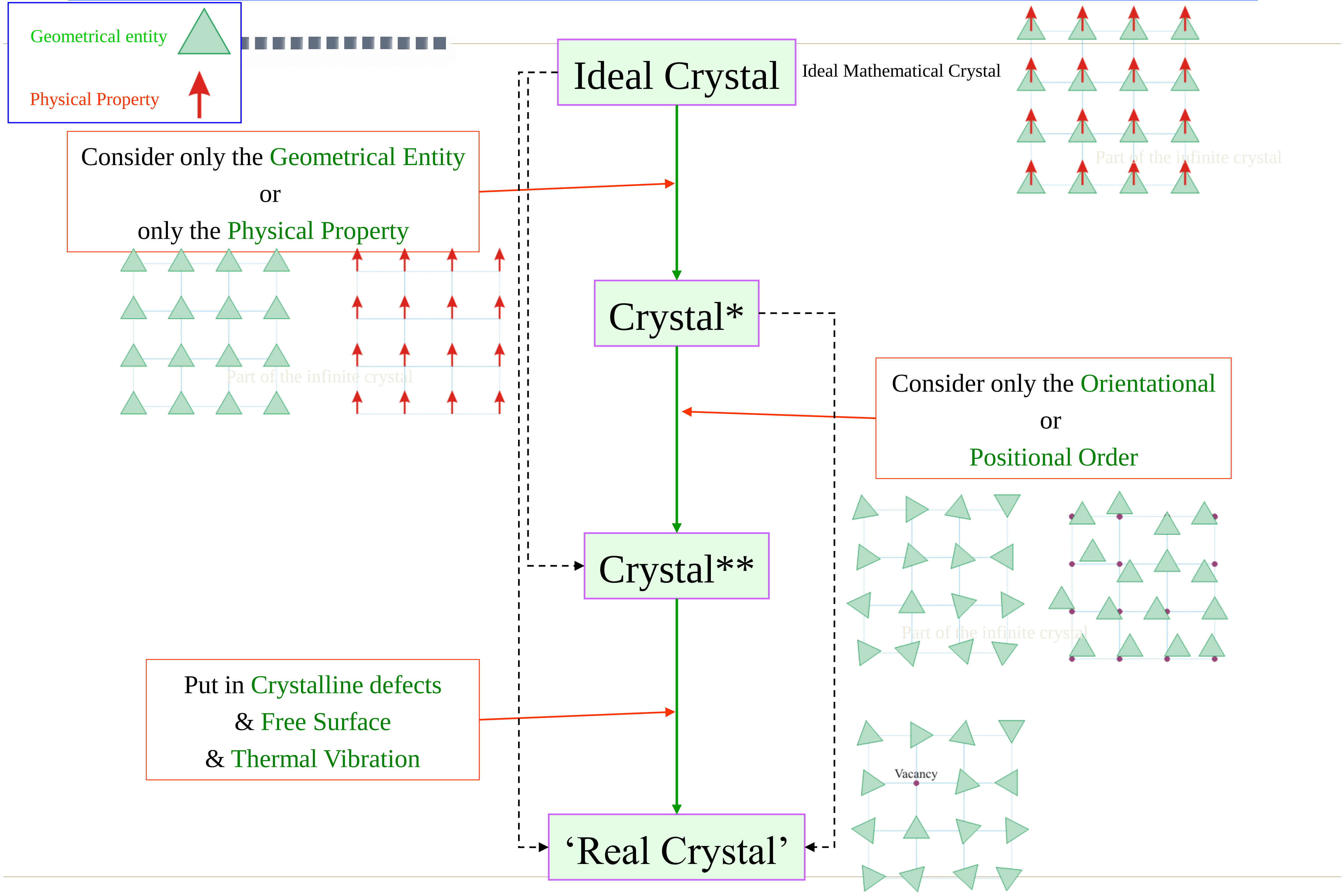
(Material) — Material or Hybrid

↓

Component

Real materials are usually complex and we start with ideal descriptions

Ideal Crystals → Real Crystals → Microstructures → Material → Component

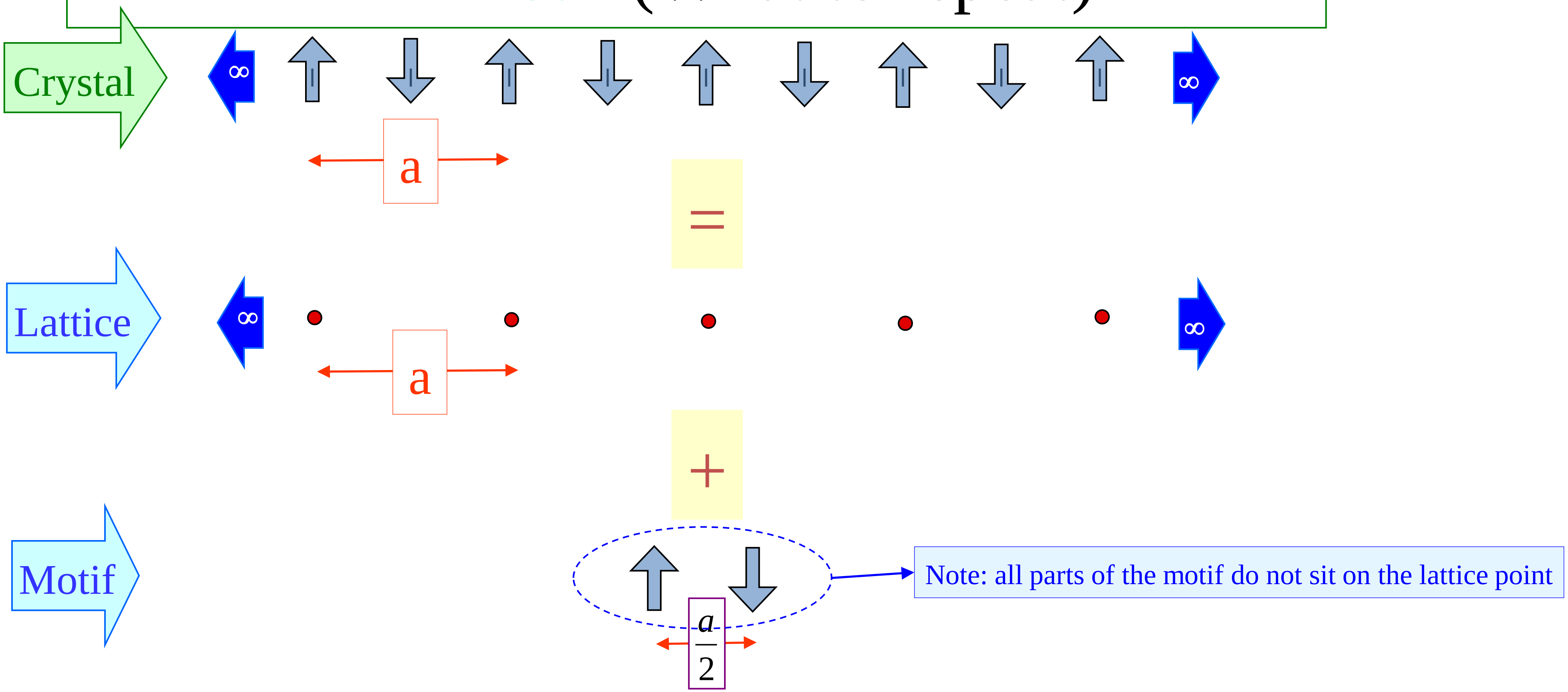


Crystal =

Lattice (Where to repeat)

+

Motif (What to repeat)

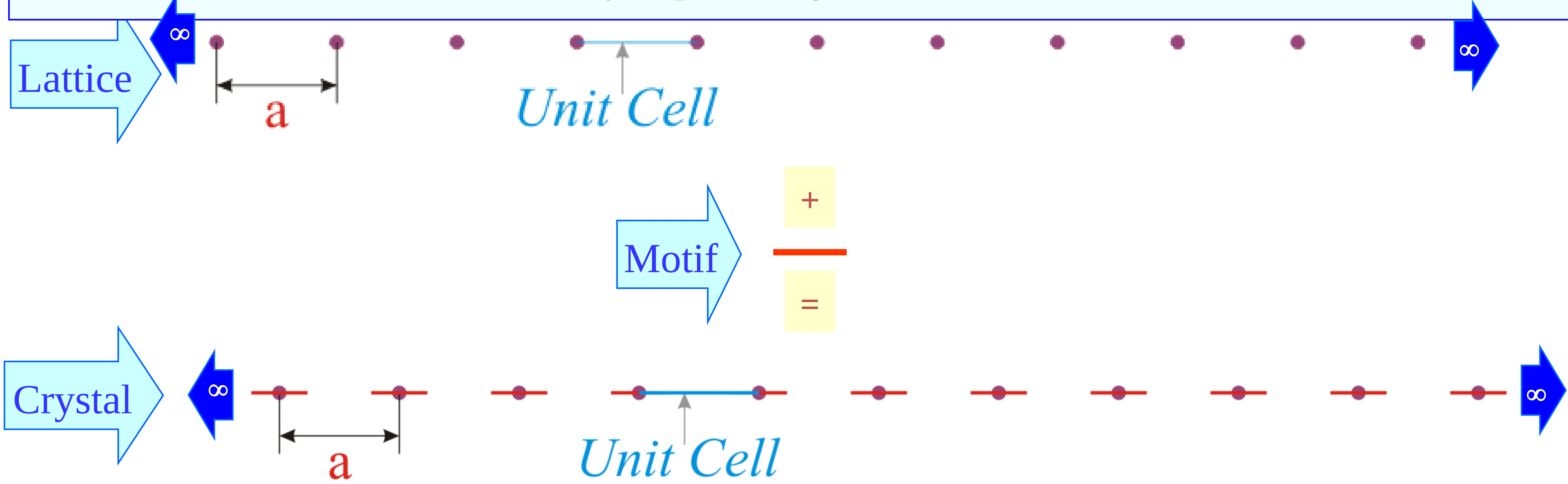


Motifs are associated with lattice points

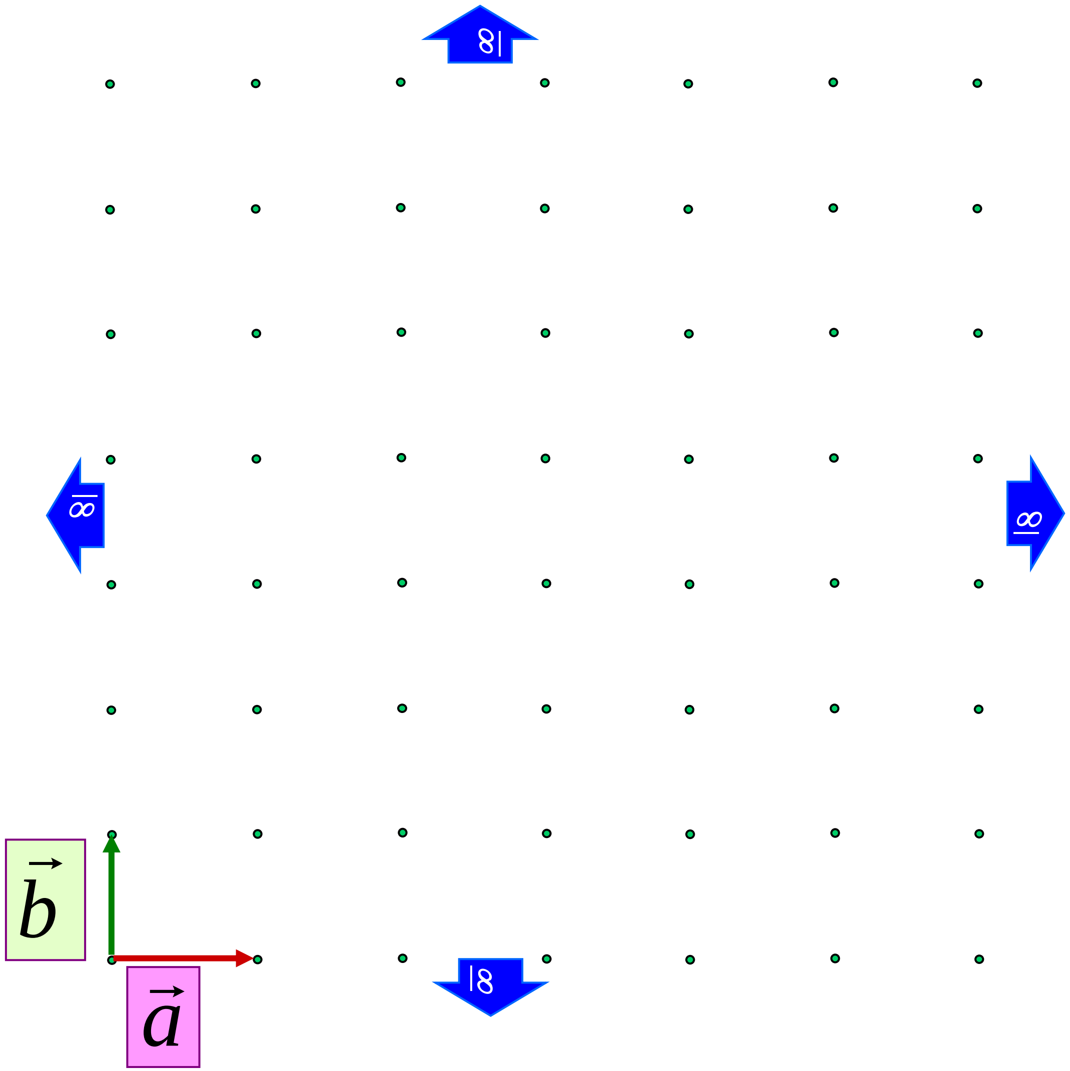
$\rightarrow$ they need NOT sit physically at the lattice point

Making a 1D Crystal

- Some of the concepts are best illustrated in lower dimensions → hence we shall construct some 1D and 2D crystals before jumping into 3D
- A strict 1D crystal = 1D lattice + 1D motif
- The only kind of 1D motif is a line segment
- An unit cell is a representative unit of the structure (finite part of an infinite structure)
  - which when translationally repeated gives the whole structure.



Lattice

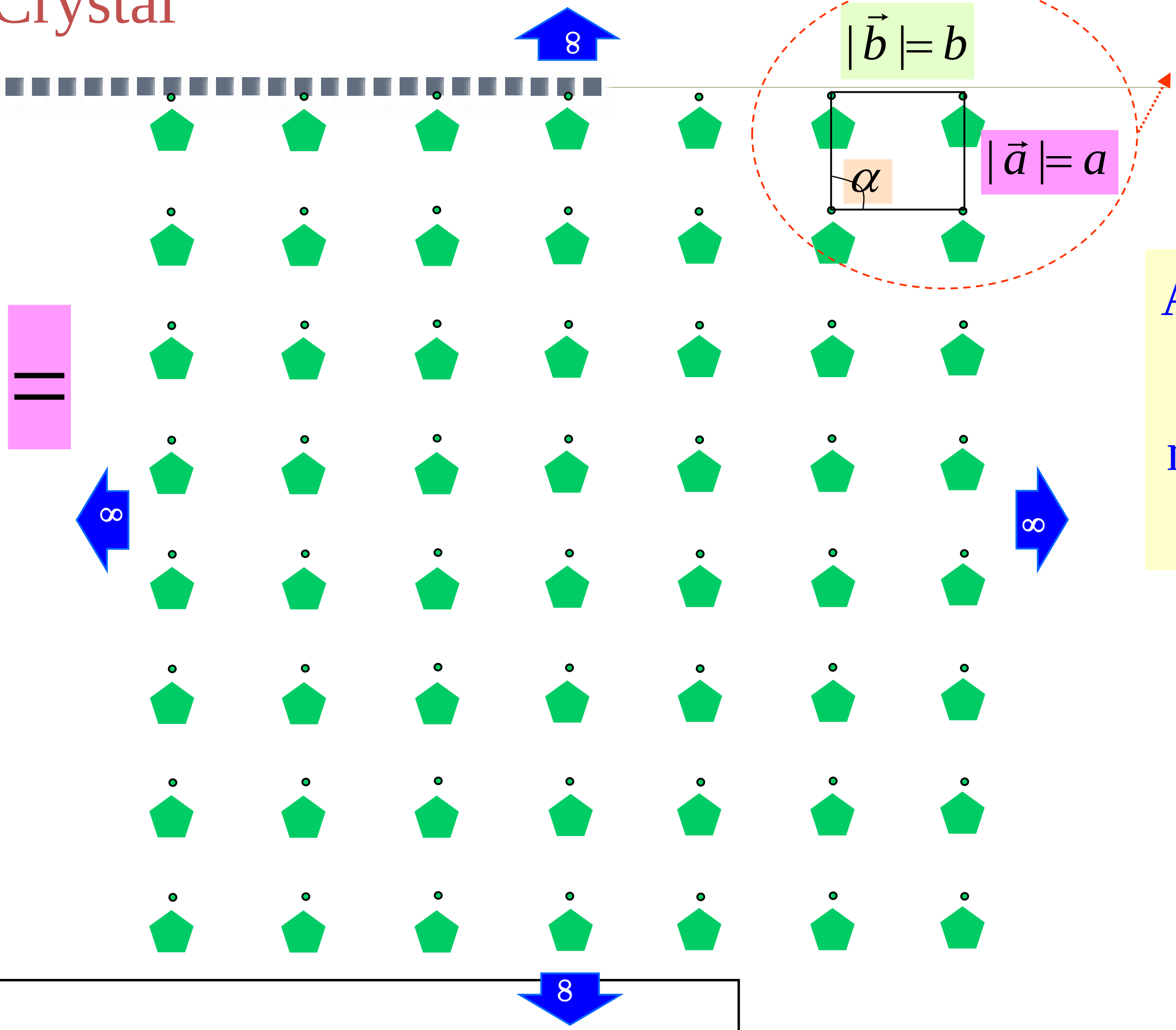


Motif

+

Continued

Crystal



The 2D crystal (& lattice) is specified using 3 lattice parameters: ‘a’, ‘b’ and α.

As before there are many ways of associating the motif with a lattice point (one of these is shown)

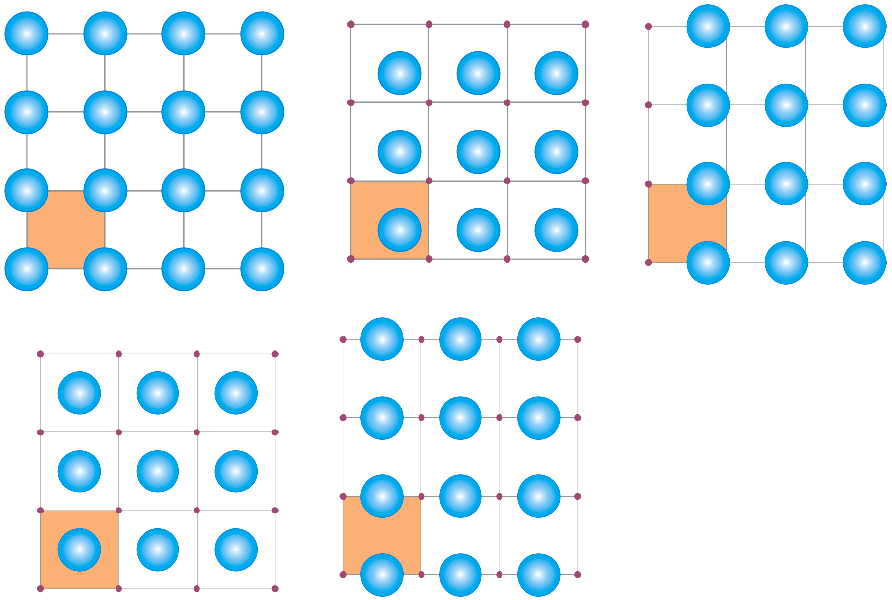
✵Note:

Each motif is identically oriented (orientationally ordered)

and

is associated exactly at the same position with each lattice point (positionally ordered)

- Motif is associated with the lattice point and need not 'sit' on the lattice point. In any case the full motif cannot 'sit' on the lattice point (just one point, maybe the centroid of an object, can 'sit' on the lattice point).
- Motif can be associated with a lattice point in many ways, as in the examples below.



In all the cases the conventional unit cell is marked in orange. This UC has lattice points as vertices.

Lattice Parameters

- The physical properties of solids depend entirely upon the arrangement of the atoms that make up the solid and the distances between them.
- The arrangement of the atoms in a crystal structure is a combination of the size and shape of the unit-cell and the arrangement of atoms inside the unit-cell.
- The shape of the unit cells is described by the lattice symmetry. Unit cell = 3-dimensional unit that repeats in space
- ***The size of the unit-cell is described in terms of its unit-cell parameters. These are the edge lengths and the angles of the unit-cell.***

- The unit cell geometry is defined in terms of six parameters*: the three edge lengths* a, b, and c, and *the three inter-axial angles* α, β, and γ indicated in Figure 3.4, and are termed the ***lattice parameters or lattice constants*** of a crystal structure
- Seven possible combinations of a, b, c & α, β, γ resulting in seven crystal systems

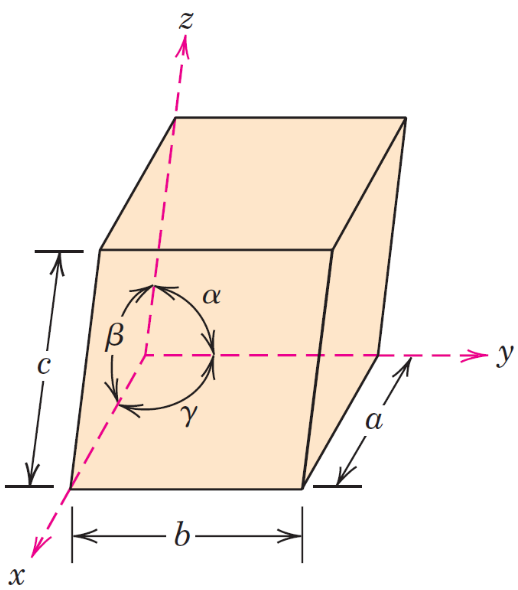
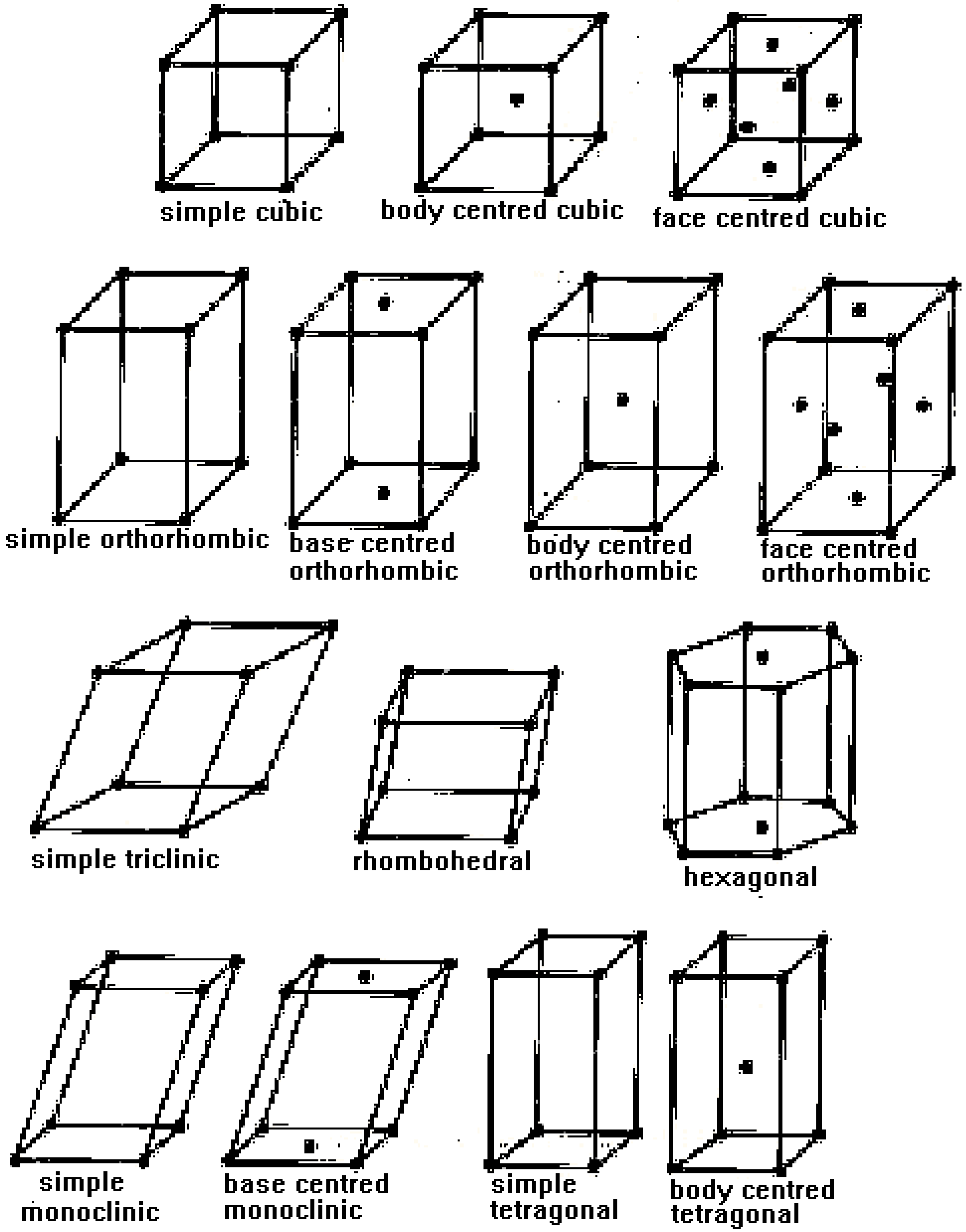


Figure 3.4 A unit cell with x, y, and z coordinate axes, showing axial lengths (a, b, and c) and interaxial angles (α, β, and γ).

Bravais Lattice

- By assigning specific values for axial lengths and inter-axial angles, unit cells of different types can be constructed
- Crystallographers have shown that only 7 different types of crystal systems are necessary to create and construct all the lattices
- Bravais showed that there are 14 possible ways of constructing the crystal lattice from seven crystal systems.
- Seven different crystal Systems
  - Cubic – 3
  - Hexagonal – 1
  - 0rthorhombic – 4
  - Tetragonal – 2
  - Rhombohedral – 1
  - Monoclinic – 2
  - Triclinic - 1

Crystal Structures - *The 14 Bravais Lattices*



In 1850, Auguste Bravais showed that crystals could be divided into 14 unit cells, which meet the following criteria.

- The unit cell is the simplest repeating unit in the crystal.
- Opposite faces of a unit cell are parallel.
- The edge of the unit cell connects equivalent points.

Crystal Structures - *The 14 Bravais Lattices*

A ***Bravais lattice*** is an *infinite array of discrete points with an arrangement and orientation that appears exactly the same viewed from any point of the array*. A three dimensional Bravais lattice consists of all points with position vectors **R** of the form:

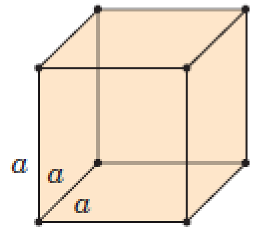
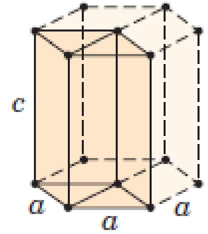
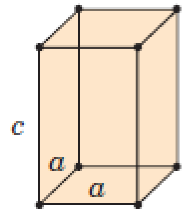
$$\mathbf{R} = \mathbf{n_1 a_1 + n_2 a_2 + n_3 a_3},$$

where the three primitive lattice vectors $\mathbf{a_i}$ are not all in the same plane, and the **n's** are integers.

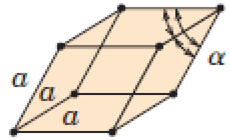
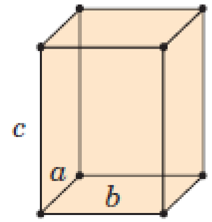
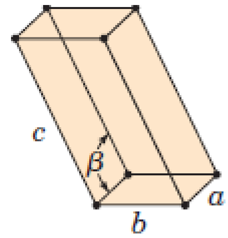
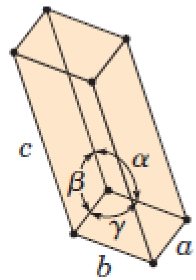
one lattice point per unit cell since the points at the eight corners are shared by eight adjacent unit cells.

Crystallographers often describe a crystal in terms of a non-primitive unit cell which is larger than the primitive cell. For instance, a body-centered cubic crystal has a non-cubic primitive cell, but it is often described in terms of a cubic conventional cell which is twice the size of the primitive cell.

Crystal Systems - Seven

	Crystal System	*Axial Relationships*	*Interaxial Angles*	*Unit Cell Geometry*
1	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	
2	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°, \gamma = 120°$	
3	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	

Crystal Systems - Seven

4	Rhombohedral (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	
5	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	
6	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90° \neq \beta$	
7	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	

Crystal Structures – Metallic Crystals

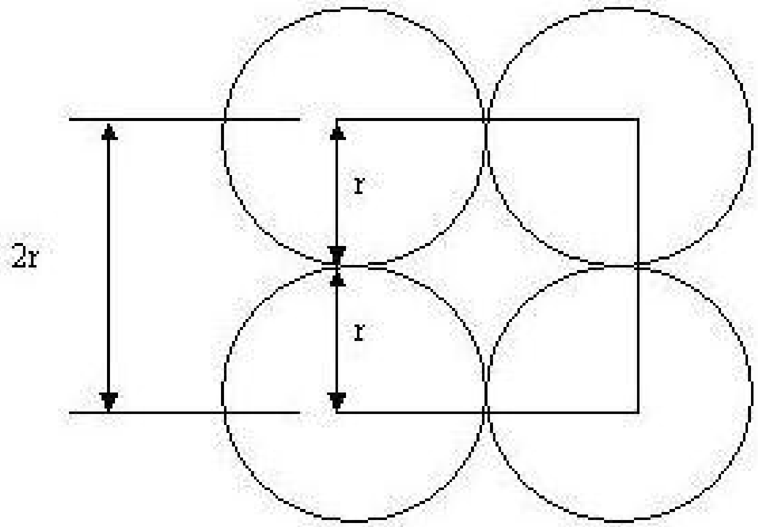
- tend to be densely packed.
- have several reasons for dense packing
  - Typically, made of heavy element.
  - Metallic bonding is not directional; i.e., no restrictions as to the number and position of nearest-neighbor atoms
  - Nearest neighbour distances tend to be small in order to lower potential energy.
- have the simplest crystal structures

- **Metallic Crystal Structures**
  - Three relatively simple crystal structures are found for most of the common metals:
    - body-centered cubic,
    - face-centered cubic, and
    - hexagonal close-packed.

Notations

- Atomic radius – It is defined as half the distance between the centres of the neighboring atoms
- Coordination Number – The no. of atoms which are directly surrounding the particular atom; it is also defined as the no. of nearest neighbors for that particular atom
- Atomic Packing fraction – Close packing of atoms in a unit cell of the crystal structure is known as APF

$$\text{APF} = \text{No. of effective atoms in the unit cell} \times \frac{\text{Volume of atoms per unit cell}}{\text{Volume of unit cell}}$$

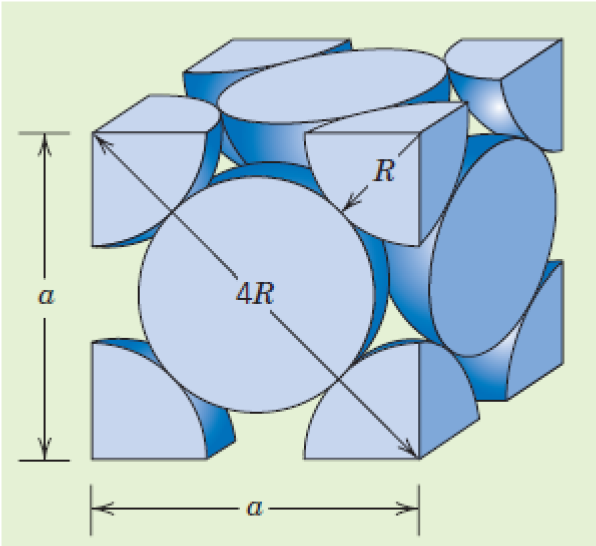


Face-centered cubic (FCC)

- Some of the familiar metals having this crystal structure are copper, aluminum, silver, and gold
- spheres or ion cores touch one another across a face diagonal; the cube edge length **a** and the atomic radius **R**

$$a = 2R\sqrt{2}$$

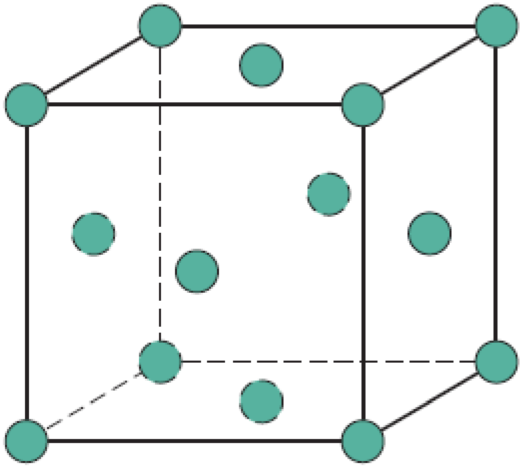
- each corner atom is shared among eight unit cells, whereas a face-centered atom belongs to only two
- Two other important characteristics of a crystal structure
  - ***coordination number , and***
  - ***the atomic packing factor (APF).***



face-diagonal the length = 4R

$$a^2 + a^2 = (4R)^2$$

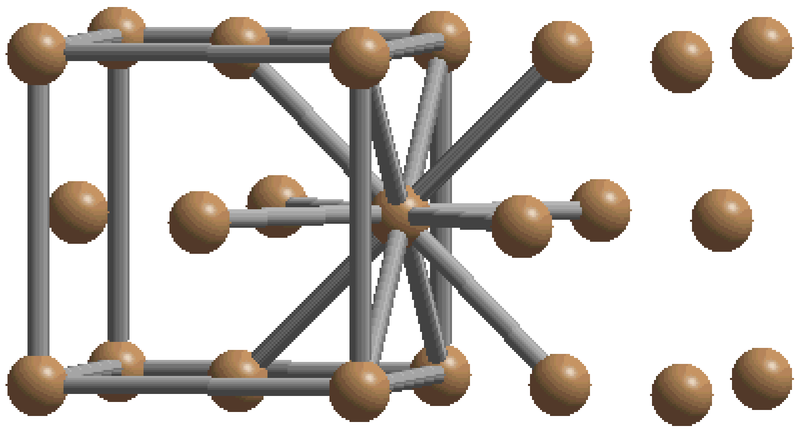
$$a = 2R\sqrt{2}$$

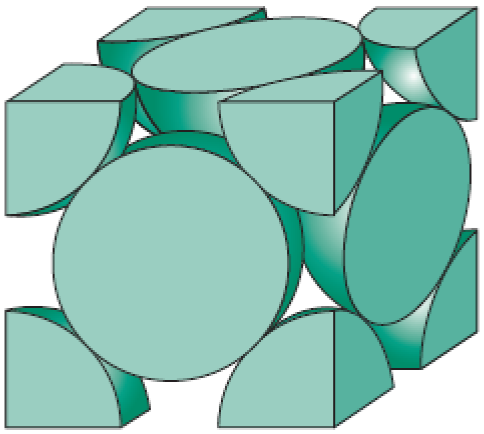


(b)

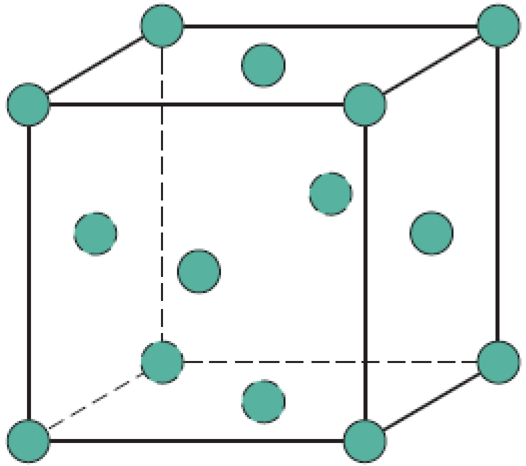
FCC - Coordination number

- each atom has the same number of nearest-neighbor or touching atoms, which is the coordination number.
- For face-centered cubics, **the coordination number is 12**.
  - the front face atom has four corner nearest-neighbor atoms surrounding it,
  - four face atoms that are in contact from behind, and
  - four other equivalent face atoms residing in the next unit cell to the front





(a)



(b)

FCC - Atomic packing factor (APF)

- The APF is the sum of the sphere volumes of all atoms within a unit cell (assuming the atomic hard-sphere model) divided by the unit cell volume

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} = \frac{V_S}{V_C}$$

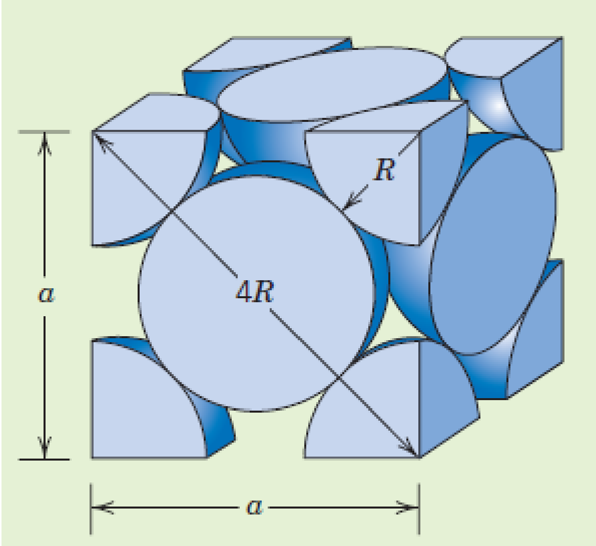
- Both the total atom and unit cell volumes may be calculated in terms of the atomic radius R

$$V_S = (4)\tfrac{4}{3}\pi R^3 = \tfrac{16}{3}\pi R^3$$

atoms per unit cell

$$\text{APF} = \frac{V_S}{V_C} = \frac{\left(\frac{16}{3}\right)\pi R^3}{16R^3\sqrt{2}} = 0.74$$

- the atomic packing factor is **0.74**.



face-diagonal the length = 4R

$$a^2 + a^2 = (4R)^2$$

$$a = 2R\sqrt{2}$$

Volume = a^3

$$V_C = a^3 = (2R\sqrt{2})^3$$

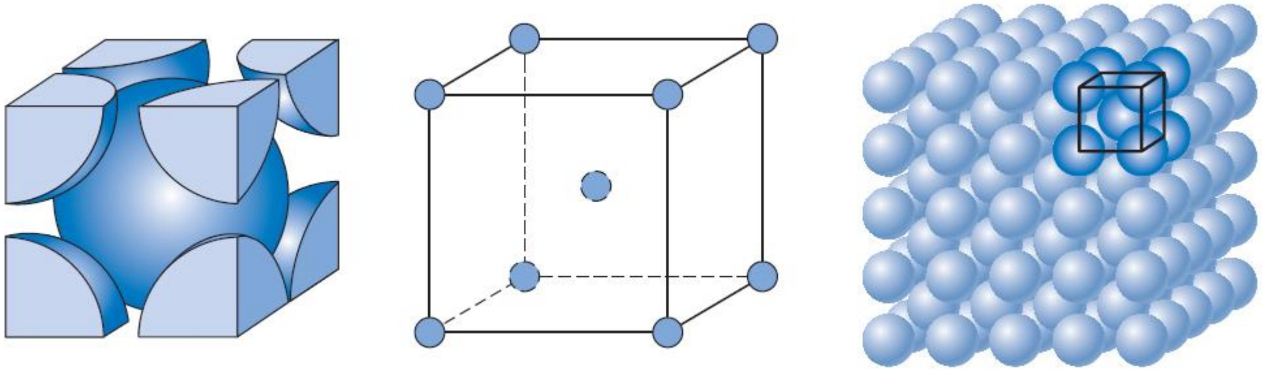
$$V_C = 16R^3\sqrt{2}$$

Comparison of crystal structures

Crystal structure	coordination #	packing factor	close packed directions
Simple Cubic (SC)	6	0.52	cube edges
Body Centered Cubic (BCC)	8	0.68	body diagonal
Face Centered Cubic (FCC)	12	0.74	face diagonal
Hexagonal Close Pack (HCP)	12	0.74	hexagonal side

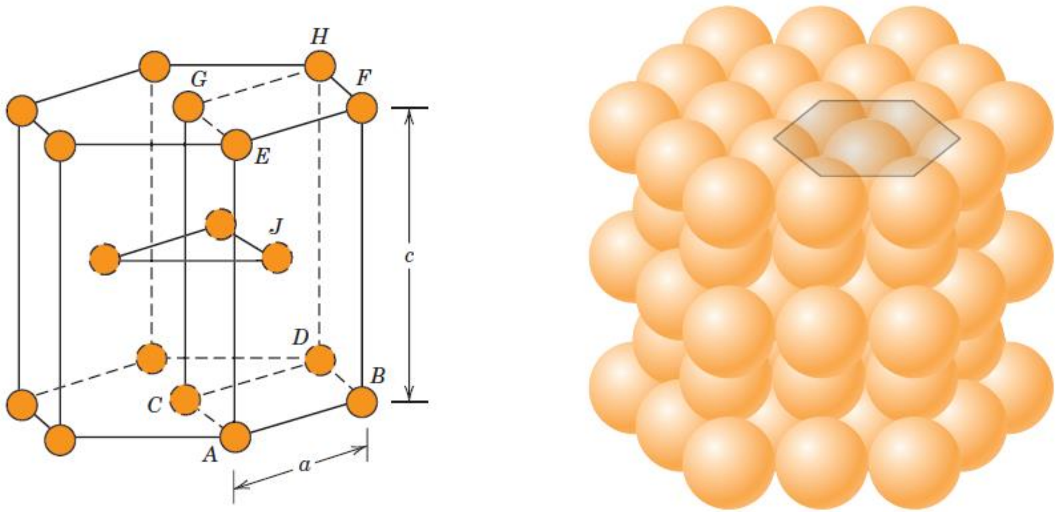
Crystal Structures

- The Body-Centered Cubic Crystal Structure



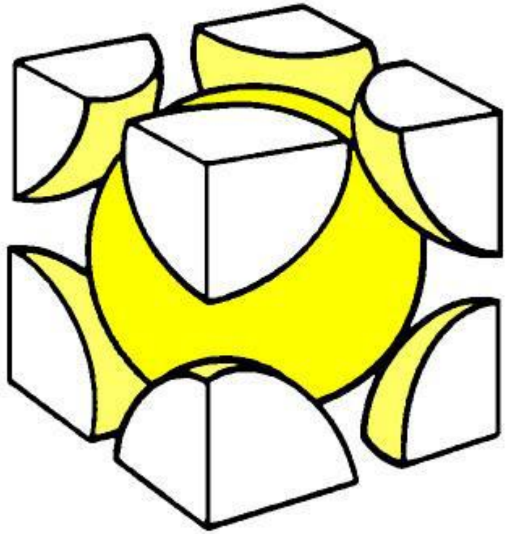
(a) (b) (c)

- The Hexagonal Close-Packed Crystal Structure



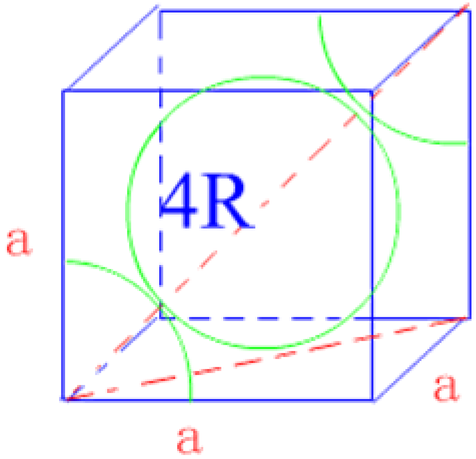
(a) (b)

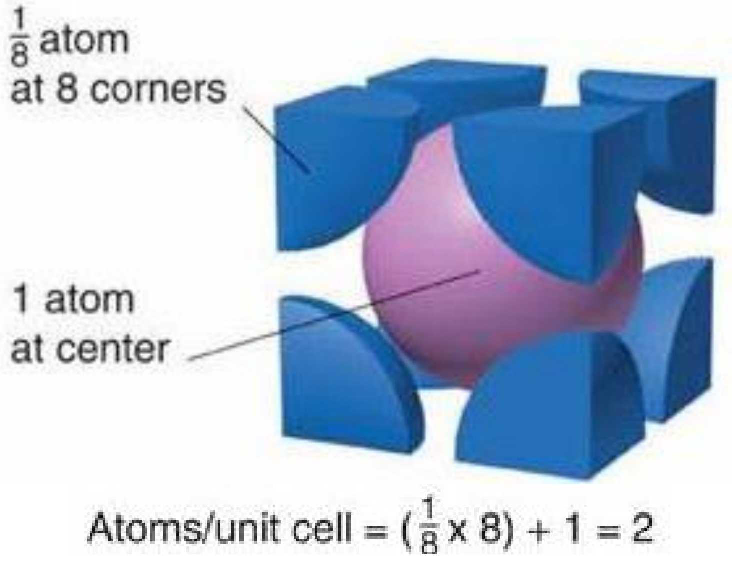
Body Centered Cubic Crystal Structure (BCC Structure)



- The hard spheres touch one another along cube diagonal
- Eight nearest neighbors surround the central atom hence **CN = 8**
- Effective Number of atoms per unit cell, n = 2
- Center atom (1) shared by no other cells: 1 x 1 = 1 and 8 corner atoms shared by eight cells: 8 x 1/8 = 1

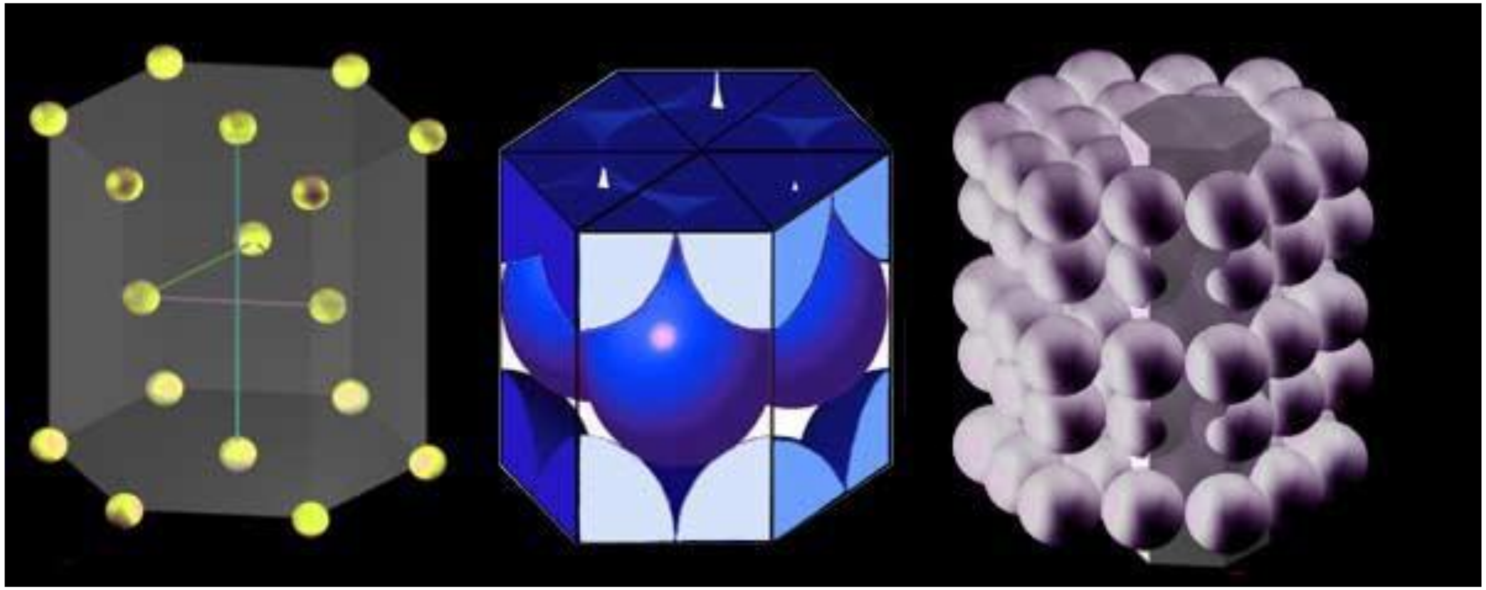
- Atomic Radius R = $\sqrt{3}\, a/4$
- APF = 0.68
- Packing Efficiency = 68%
- Some of the materials that possess BCC structure include lithium, sodium, potassium, barium, vanadium, alpha-iron and tungsten

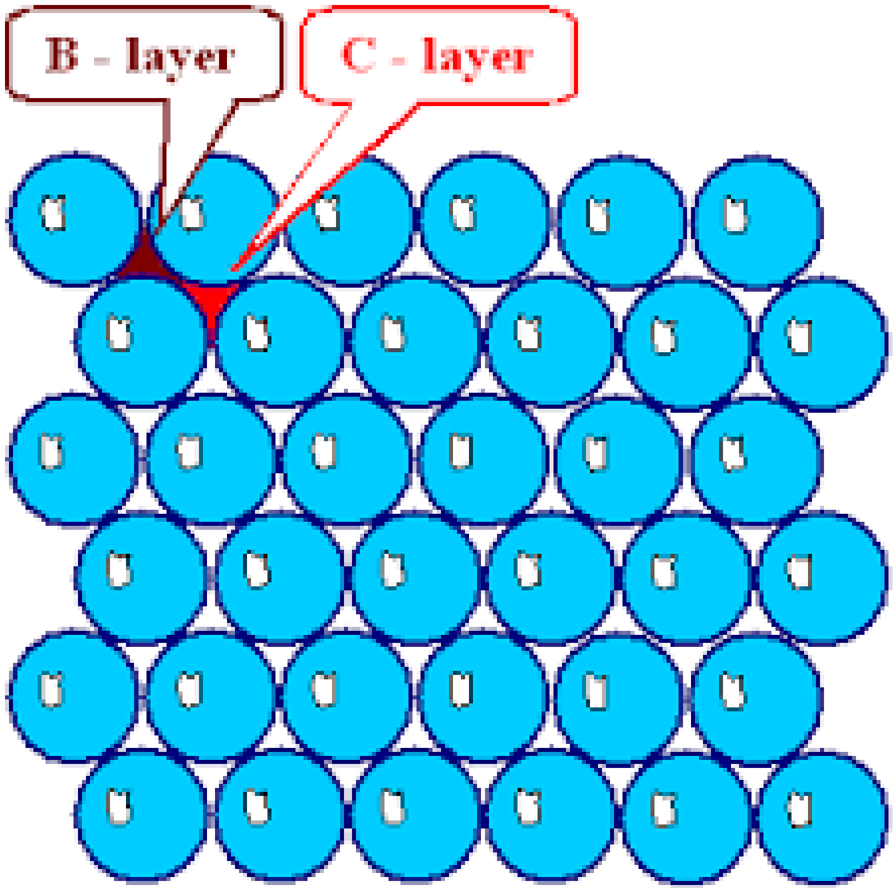


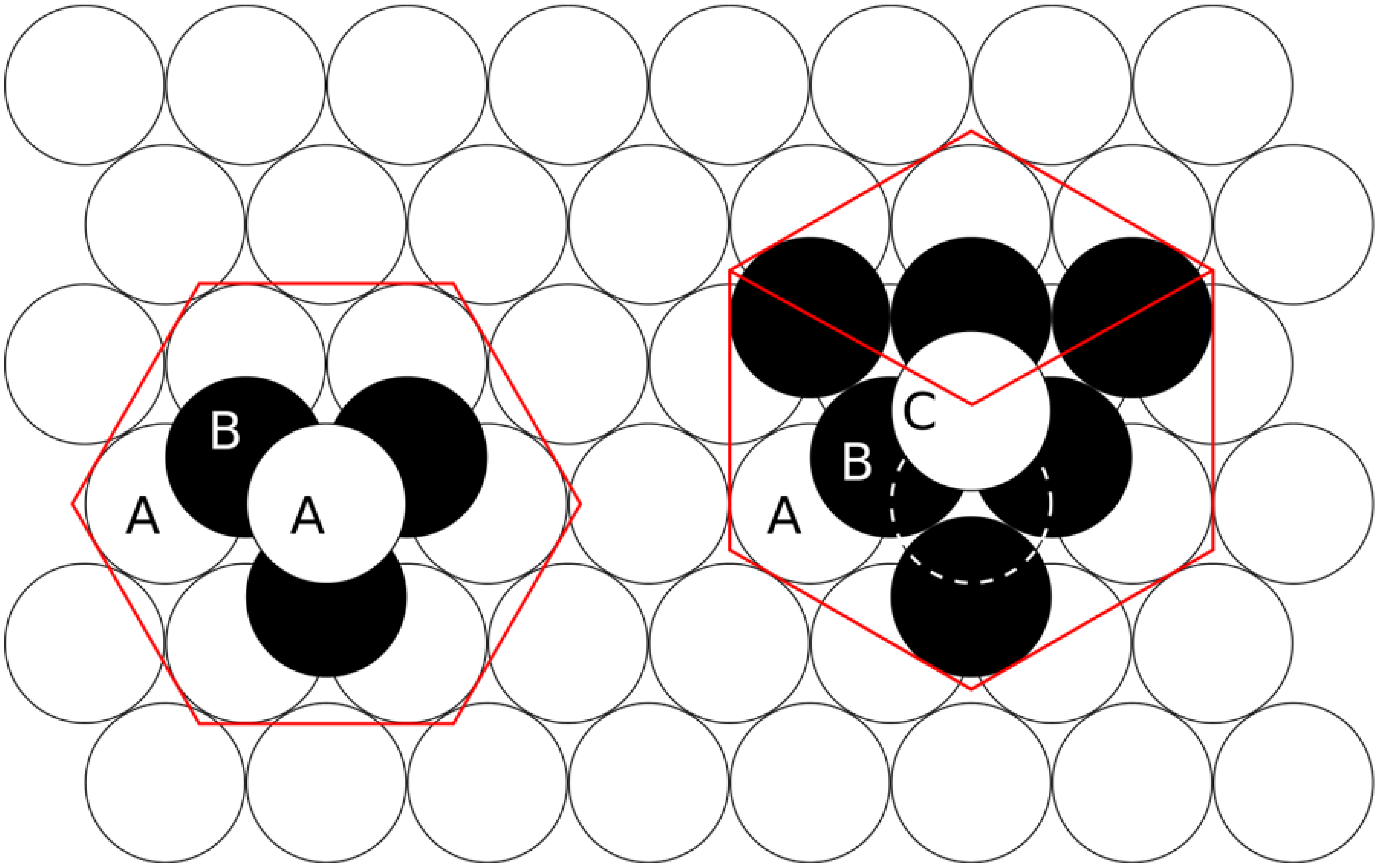


Hexagonal Close Pack Structure

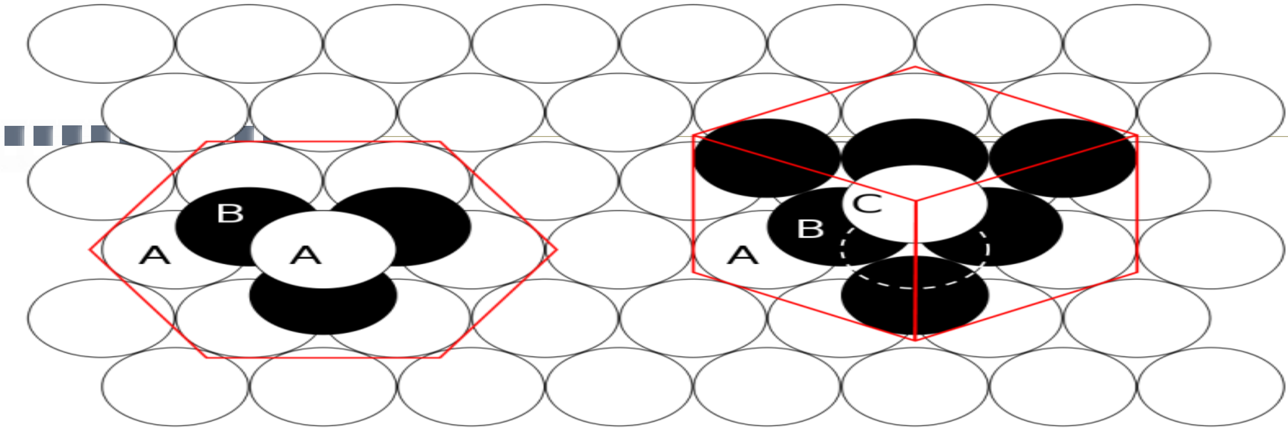
- Metals don't crystallize into simple hexagonal structure since the APF is too low
- The atoms in the hcp structure are also packed along close-packed planes.
- It should also be noted that both the FCC and HCP structures are known as close-packed structures with crystallographic planes having the same arrangement of atoms;
- However, the order of stacking the planes is different. Atoms in the hexagonal close-packed planes (called the basal planes) have the same arrangement as those in the FCC close-packed planes

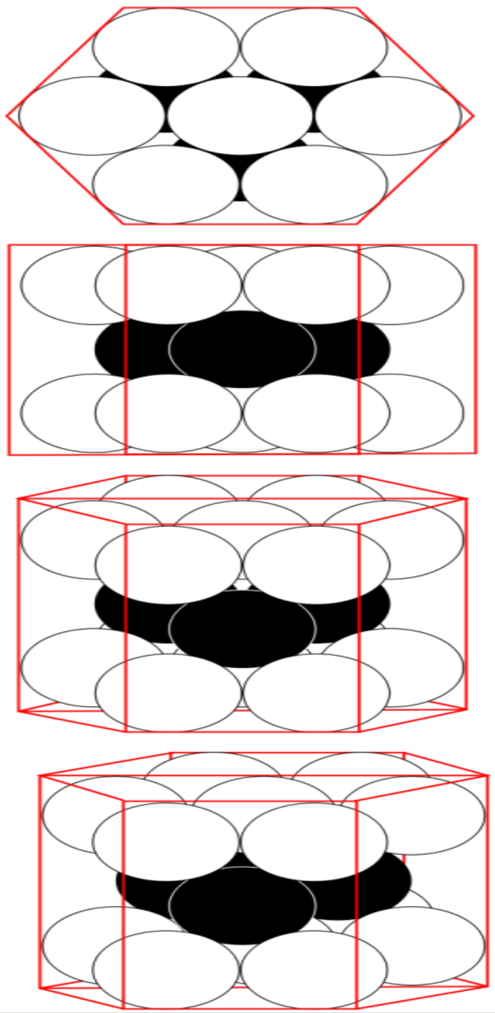


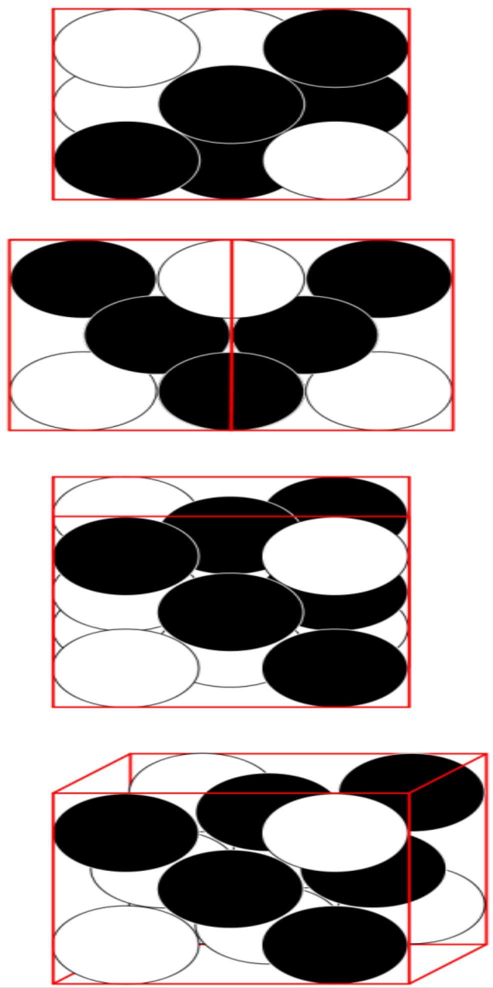

B - layer
C - layer



B
A
A
C
B
A

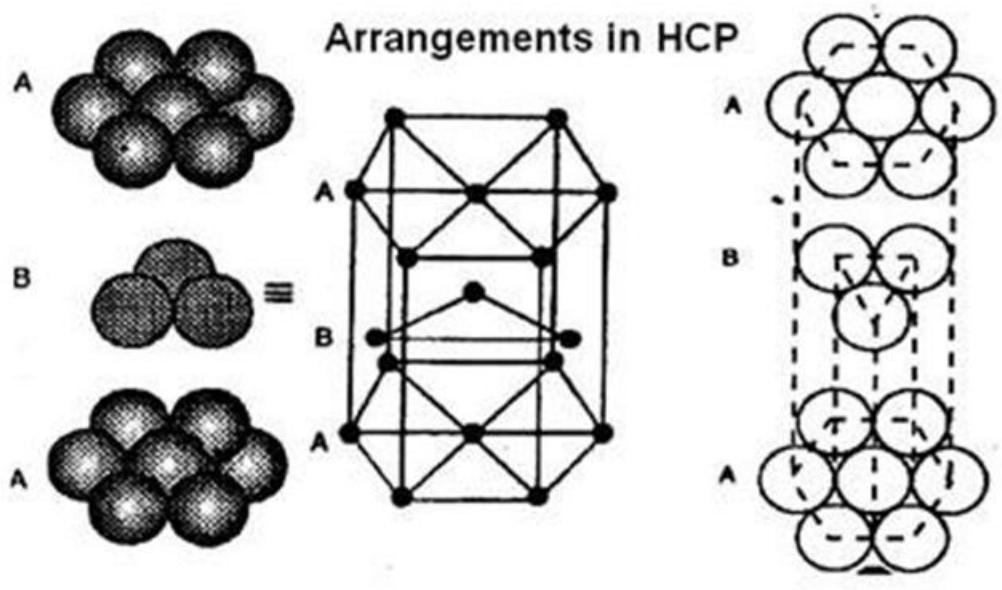

B
A
A
C
B
A

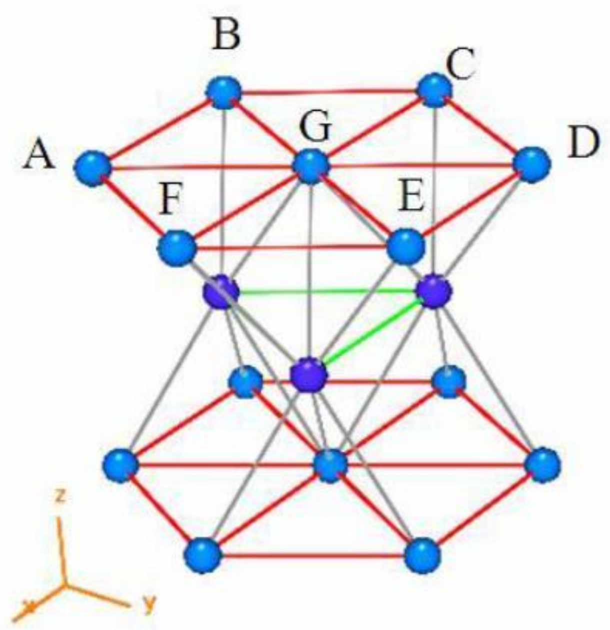




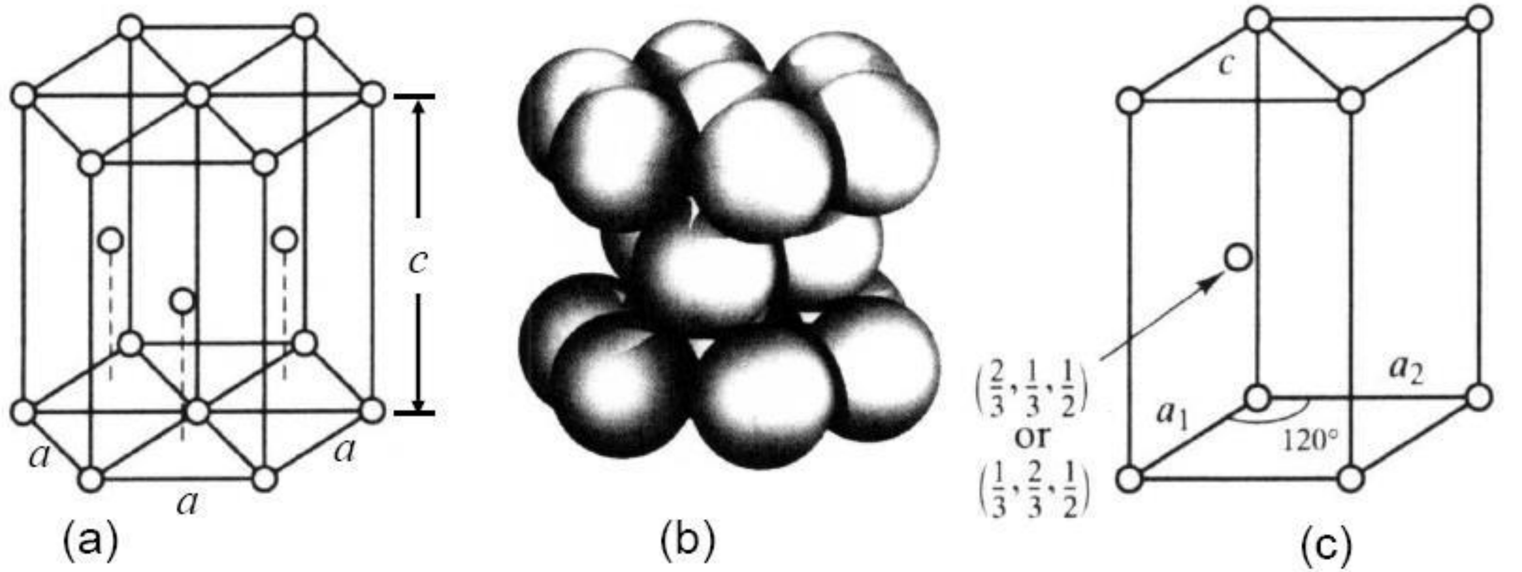
Hexagonal Close Pack Structure

- The coordination number, CN = 12 (same as in FCC)
- Atomic radius = a/2
- Number of atoms per unit cell, n = 6.
  - – 3 mid-plane atoms shared by no other cells: 3 x 1 = 3
  - – 12 hexagonal corner atoms shared by 6 cells: 12 x 1/6 = 2
  - – 2 top/bottom plane center atoms shared by 2 cells: 2 x 1/2 = 1



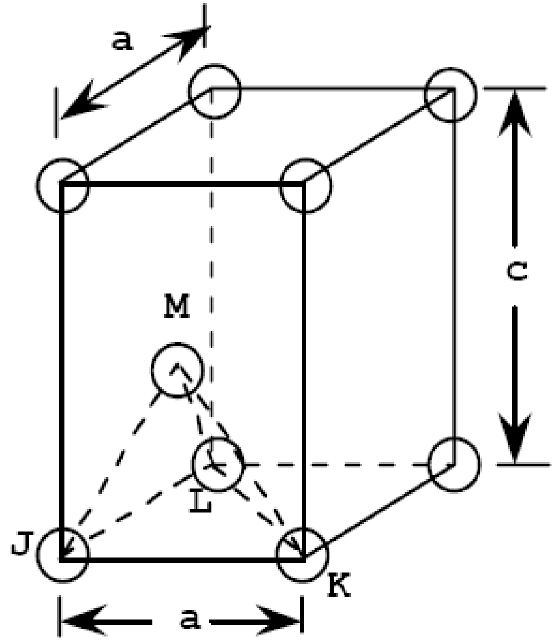


Expression for ideal c/a ratio - HCP

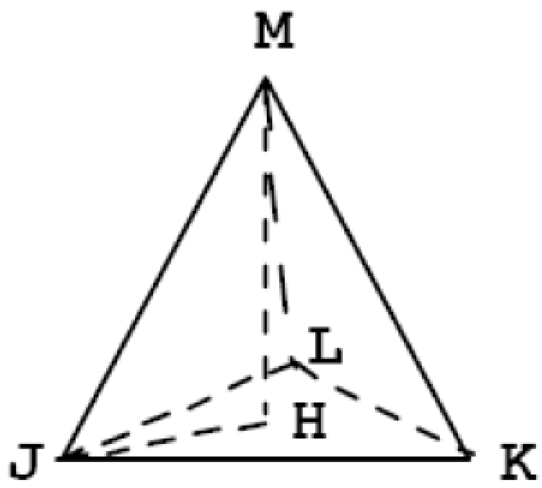


The HCP crystal structure: (a) 3 unit cells, (b) hard sphere model and (c) the unit cell

- The atom at point M is midway between the top and bottom faces of the unit cell that is MH= c/2.
- And, since atoms at points J, K, and M, all touch one another



Expression for ideal c/a ratio - HCP



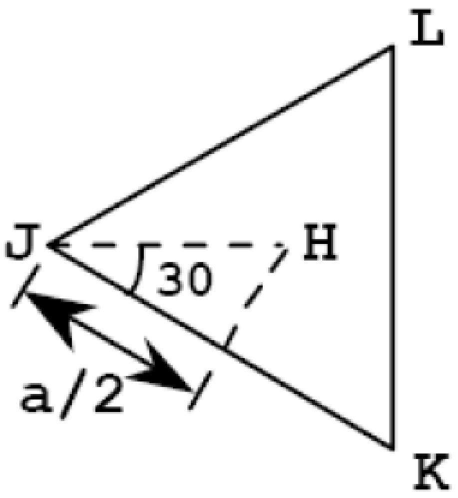
$JM = JK = 2R = a$
where **R is the atomic radius.**

- Furthermore, from triangle JHM,

$$(JM)^2 = (JH)^2 + (MH)^2$$

$$a^2 = (JH)^2 + (c/2)^2$$

Now, we can determine the JH length by consideration of triangle JKL, which is an equilateral triangle

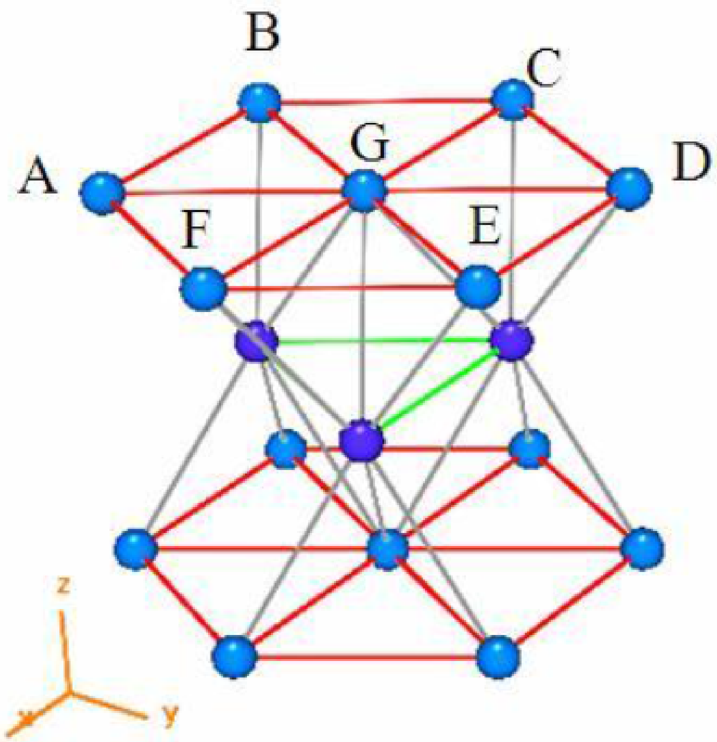


$JH = a/\sqrt{3}$

Substituting this in the above expression yields c/a ratio as 1.633

HCP Structure - Atomic packing factor

APF



- Δ BCG,

Area = ½ * base * height

= ½ * a * a Sin60°

<BCG = BGC = 60°

Atomic Radius = a/2

Total Area ABCDEFG = 6 * ½ * a^2 Sin60°

Vol. of HCP Unit Cell = Area * height

= 3 a^2c Sin60°

APF = 6* 4/3 Π r3 / 3 a^2c Sin60°

(Replace r = a/2; c/a=1.633) ----- APF = 0.74

Examples for HCP : Cd, Mg, Zn, Ti have this crystal structure

Crystal Structures - Density computations

Theoretical density (ρ) for metals

$$Density(\rho) = \frac{mass(m)}{volume(v)}$$

mass = (number of atoms per unit cell) x (mass of each atom)

mass of each atom = atomic weight/Avogadro's number

$$\rho = \frac{nA}{V_C N_A}$$

n = number of atoms associated with each unit cell

A = atomic weight (g/mol)

V_C *= volume of the unit cell (cm^3 per unit cell)*

N_A *= Avogadro's number (6.022 x 10^{23} atoms/mol)*

Theoretical Density Computation for Copper

Question: Copper has an atomic radius of 0.128 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Compute its theoretical density and compare the answer with its measured density.

Solution : the crystal structure is FCC, n, the number of atoms per unit cell, is 4. Furthermore, the atomic weight A_{Cu} is given as 63.5 g/mol.

- The unit cell volume V_C for FCC is $16R^3\sqrt{2}$ where R, the atomic radius, is 0.128 nm.

Substitution for the various parameters into Equation 3.5 yields

$$\rho = \frac{nA_{\mathrm{Cu}}}{V_C N_{\mathrm{A}}} = \frac{nA_{\mathrm{Cu}}}{(16R^3\sqrt{2})N_{\mathrm{A}}}$$

$$= \frac{(4\ \text{atoms/unit cell})(63.5\ \text{g/mol})}{[16\sqrt{2}(1.28 \times 10^{-8}\ \text{cm})^3/\text{unit cell}](6.022 \times 10^{23}\ \text{atoms/mol})}$$

$$= 8.89\ \text{g/cm}^3$$

The literature value for the density of copper is 8.94 g/cm^3, which is in very close agreement with the foregoing result.

Polymorphism and Allotropy

- Polymorphism
  - Same compound occurring in more than one crystal structure
  - depends on both the temperature and the external pressure
  - Eg. calcite, aragonite and vaterite minerals , different forms of calcium carbonate
- Allotropy
  - Polymorphism in elemental solids (e.g., carbon)

Crystallographic Points, Directions, and Planes

- When dealing with crystalline materials, it often becomes necessary *to specify a particular point within a unit cell, a crystallographic direction, or some crystallographic plane of atoms.*
- Labeling conventions have been established in which *three numbers or indices* are used to designate ***point locations, directions, and planes***.
- The basis for determining index values is the unit cell, with a right-handed coordinate system consisting of three (x, y, and z) axes situated at one of the corners and coinciding with the unit cell edges, as shown in Figure.
- For some crystal systems—namely, hexagonal, rhombohedral, monoclinic, and triclinic-

  -the three axes are not mutually perpendicular.

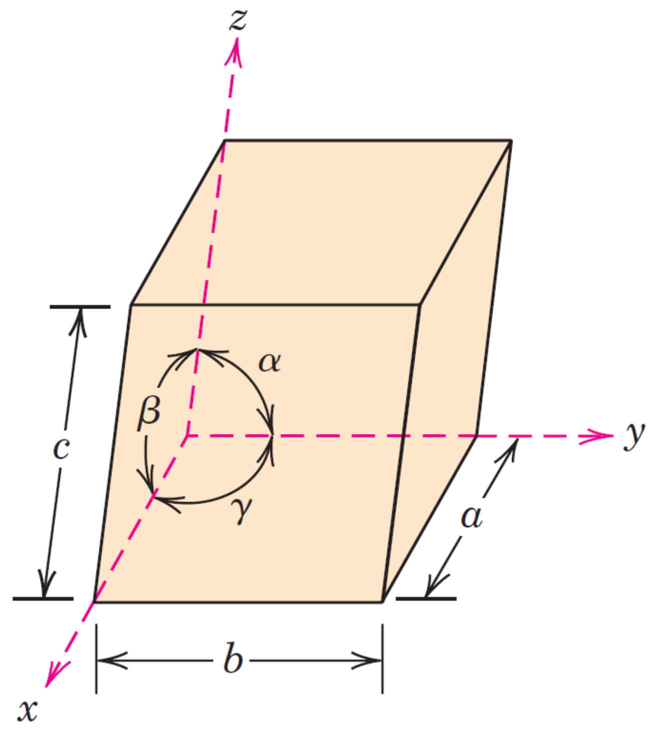
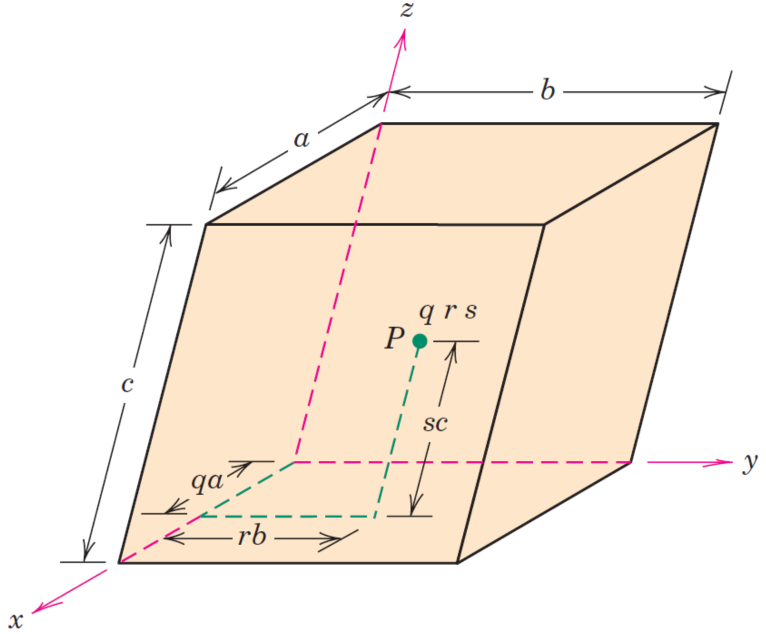


Figure 3.4 A unit cell with x, y, and z coordinate axes, showing axial lengths (a, b, and c) and interaxial angles (α, β, and γ).

Point Coordinates

- The position of any point located within a unit cell may be specified in terms of its coordinates as fractional multiples of the unit cell edge lengths (i.e., in terms of a, b, and c
- The position of any point located within a unit cell may be specified in terms of its coordinates as fractional multiples of the unit cell edge lengths (i.e., in terms of a, b, and c).
- ***To illustrate, consider the unit cell and the point P situated therein as shown in Figure.***
- We specify the position of P in terms of the generalized coordinates q, r, and s

  where q is some fractional length of a along the x axis, r is some fractional length of b along the y axis, and s is some fractional length of c along the z axis.



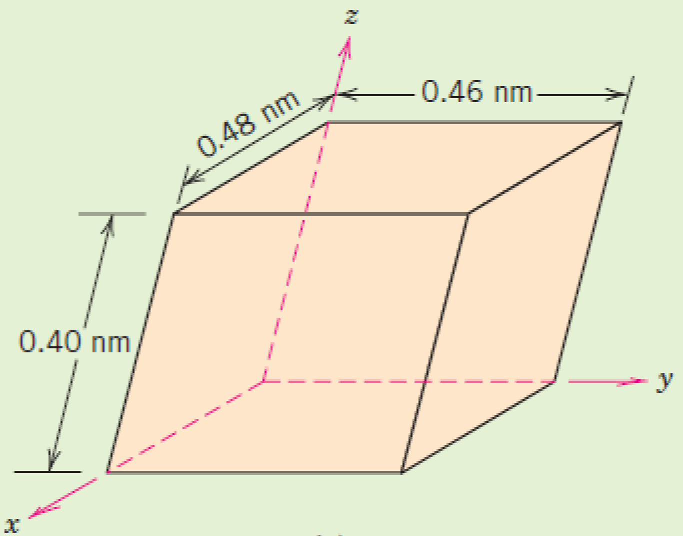
- Thus, the position of P is designated using coordinates q r s with values that are less than or equal to unity. Furthermore, we have chosen not to separate these coordinates by commas or any other punctuation marks (which is the normal convention).

Point Coordinates

EXAMPLE PROBLEM 3.4

Location of Point Having Specified Coordinates

For the unit cell shown in the accompanying sketch (*a*), locate the point having coordinates $\frac{1}{4}\ 1\ \frac{1}{2}$.

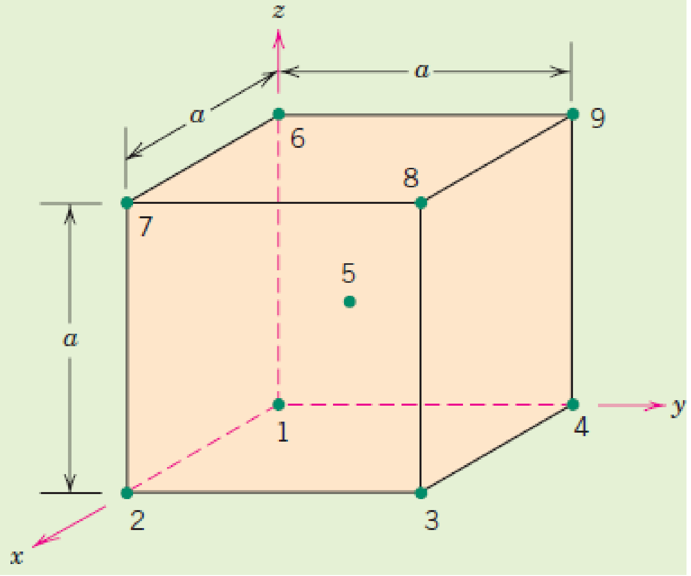


(*a*)

Point Coordinates

Specification of Point Coordinates

- Specify point coordinates for all atom positions for a BCC unit cell



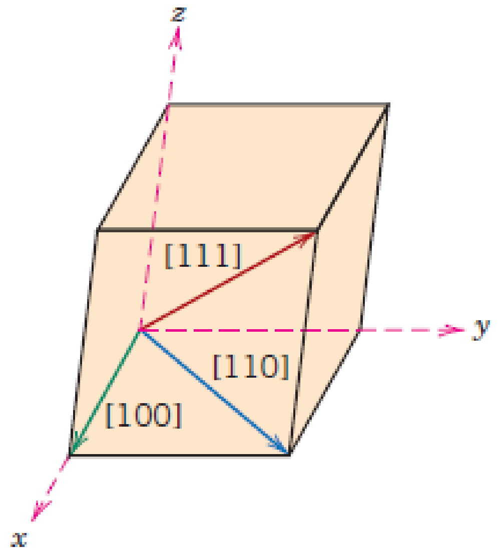
Point Number	Fractional Lengths			Point Coordinates
	x axis	y axis	z axis	
1				
2				
3				
4				
5				
6				
7				
8				
9				

Crystallographic Directions

- ***A crystallographic direction is defined as a line between two points, or a vector***.

Steps used to determine the three directional indices:

1. A vector of convenient length is *positioned such that it passes through the origin* of the coordinate system. Any vector may be translated throughout the crystal lattice without alteration, if parallelism is maintained.
2. The length of the vector projection on each of the three axes is determined; these are measured in terms of the unit cell dimensions a, b, and c.
3. These three numbers are multiplied or divided by a common factor to reduce them to the smallest integer values.
4. The three indices, not separated by commas, are enclosed in square brackets, thus: [uvw]. The u, v, and w integers correspond to the reduced projections along the x, y, and z axes, respectively.



The [100], [110], and [111] directions within a unit cell.

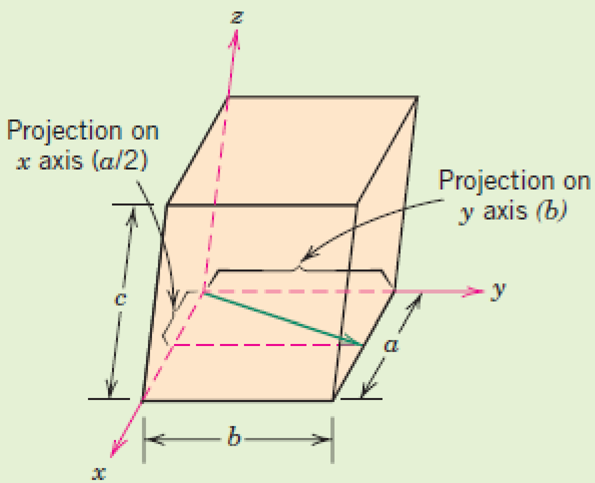
For each of the three axes, there will exist both ***positive and negative*** coordinates. Thus negative indices are also possible, which are represented by a bar over the appropriate index. Eg. the [$1\bar{1}1$]

Crystallographic Directions

EXAMPLE PROBLEM 3.6

Determination of Directional Indices

Determine the indices for the direction shown in the accompanying figure.



	x	y	z
Projections	$a/2$	b	$0c$
Projections (in terms of a, b, and c)	$\frac{1}{2}$	1	0
Reduction	1	2	0
Enclosure		[120]	

Crystallographic Directions

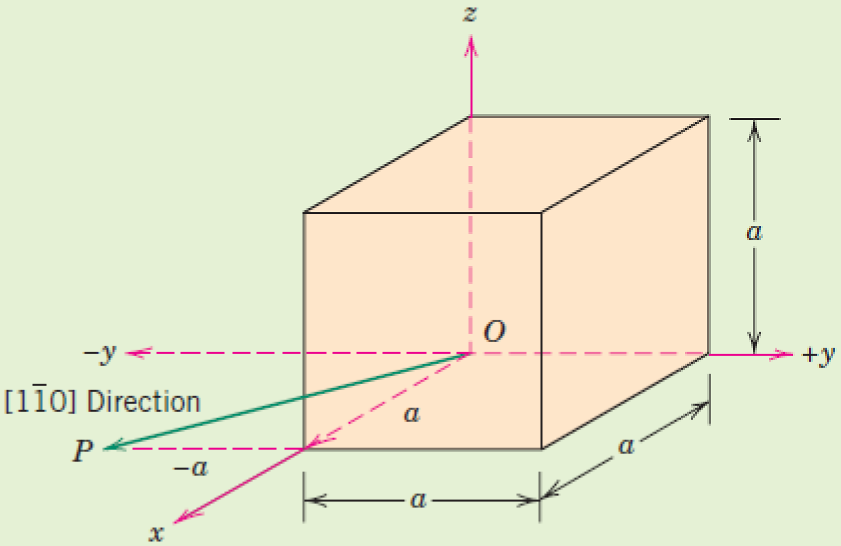
EXAMPLE PROBLEM 3.7

Construction of Specified Crystallographic Direction

Draw a $[1\bar{1}0]$ direction within a cubic unit cell.

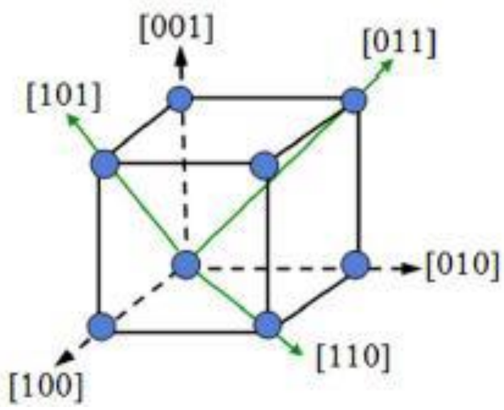
Solution

First construct an appropriate unit cell and coordinate axes system. In the accompanying figure the unit cell is cubic, and the origin of the coordinate system, point O, is located at one of the cube corners.

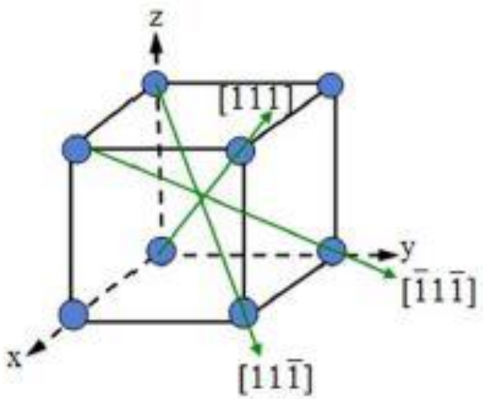


Crystallographic Directions - Family

- For some crystal structures, several nonparallel directions with different indices are crystallographically equivalent; this means that the spacing of atoms along each direction is the same.
- The directions in a crystal are given by specifying the coordinates (u, v, w) of a point on a vector passing through the origin. It is indicated as [uvw]. For example, the direction [110] lies on a vector whose projection lengths on x and y axes are one unit.
- Directions of a form or family like [110], [101], [011] are written as <110>



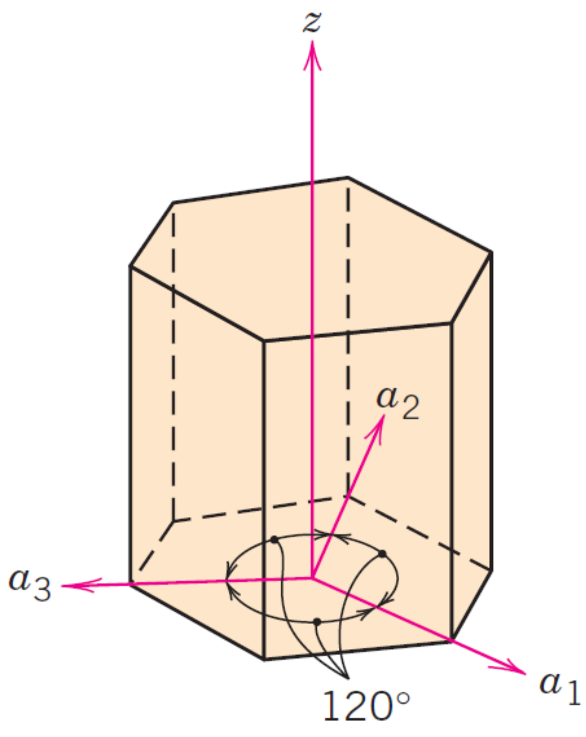
<100> and <110> family



<111> family

Hexagonal Crystals

- **Problem** – In crystals having hexagonal symmetry, some crystallographic equivalent directions will not have the same set of indices.
- This is circumvented by utilizing ***a four-axis, or Miller–Bravais, coordinate system*** as shown in Figure.
- The three a1, a2, and a3 axes are all contained within a single plane (called the basal plane) and are at 120° angles to one another.
- The z axis is perpendicular to this basal plane.
- Directional indices, will be denoted by four indices, as [uvtw]; by convention, the first three indices pertain to projections along the respective a1, a2, and a3 axes in the basal plane



Coordinate axis system for a hexagonal unit cell (Miller–Bravais scheme).

Hexagonal Crystals

Conversion from the three-index system to the four-index system,

$$[u'v'w'] \rightarrow [uvtw]$$

is accomplished by the following formulas:

$$u = \frac{1}{3}(2u' - v')$$

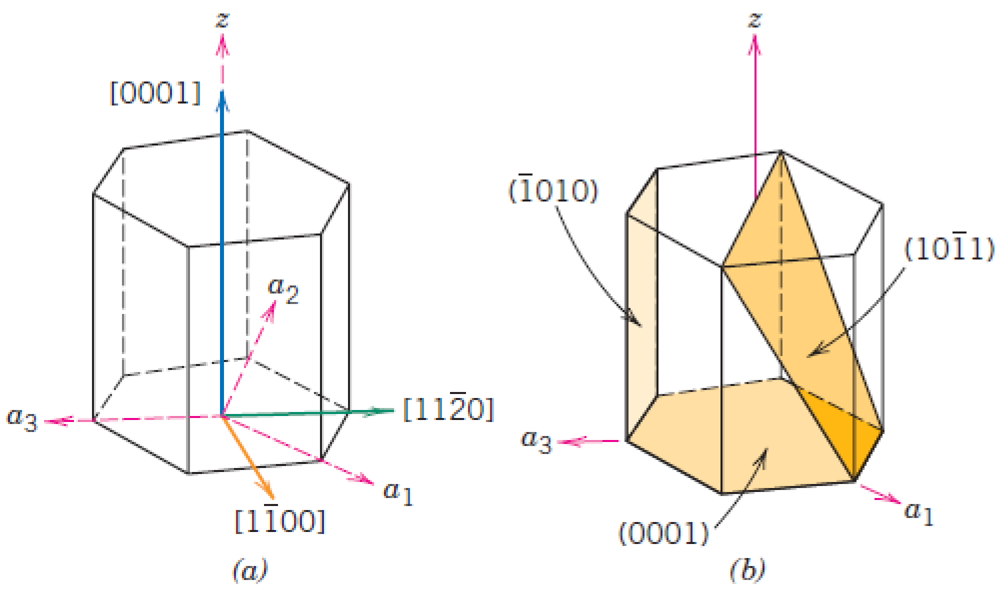
$$v = \frac{1}{3}(2v' - u')$$

$$t = -(u + v)$$

$$w = w'$$

Eg. the [010] direction becomes $[\bar{1}2\bar{1}0]$.

Several different directions are indicated in the hexagonal unit cell



Conversion of Directional Indices for a Hexagonal Unit Cell

Convert the [111] direction into the four-index system for hexagonal crystals

Solution

$$u' = 1; v' = 1; w' = 1$$

Thus,

$$u = \frac{1}{3}(2u' - v') = \frac{1}{3}[(2)(1) - 1] = \frac{1}{3}$$

$$v = \frac{1}{3}(2v' - u') = \frac{1}{3}[(2)(1) - 1] = \frac{1}{3}$$

$$t = -(u + v) = -\left(\frac{1}{3} + \frac{1}{3}\right) = -\frac{2}{3}$$

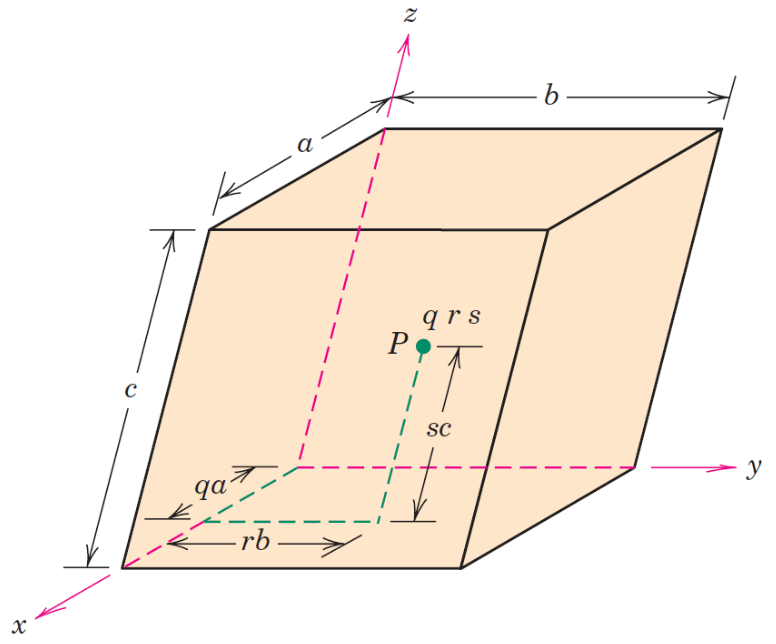
$$w = w' = 1$$

Multiplication of the preceding indices by 3 reduces them to the lowest set, which yields values for u, v, t, and w of 1, 1, 2 and 3, respectively.

Hence, the [111] direction becomes [$11\bar{2}3$].

Crystallographic Planes

- The orientations of planes for a crystal structure are represented in a similar manner.
- Again, the unit cell is the basis, with the three-axis coordinate system as represented in Figure. In all but the hexagonal crystal system, crystallographic planes are specified by *three Miller indices* as (hkl).
- Any two planes parallel to each other are equivalent and have identical indices.

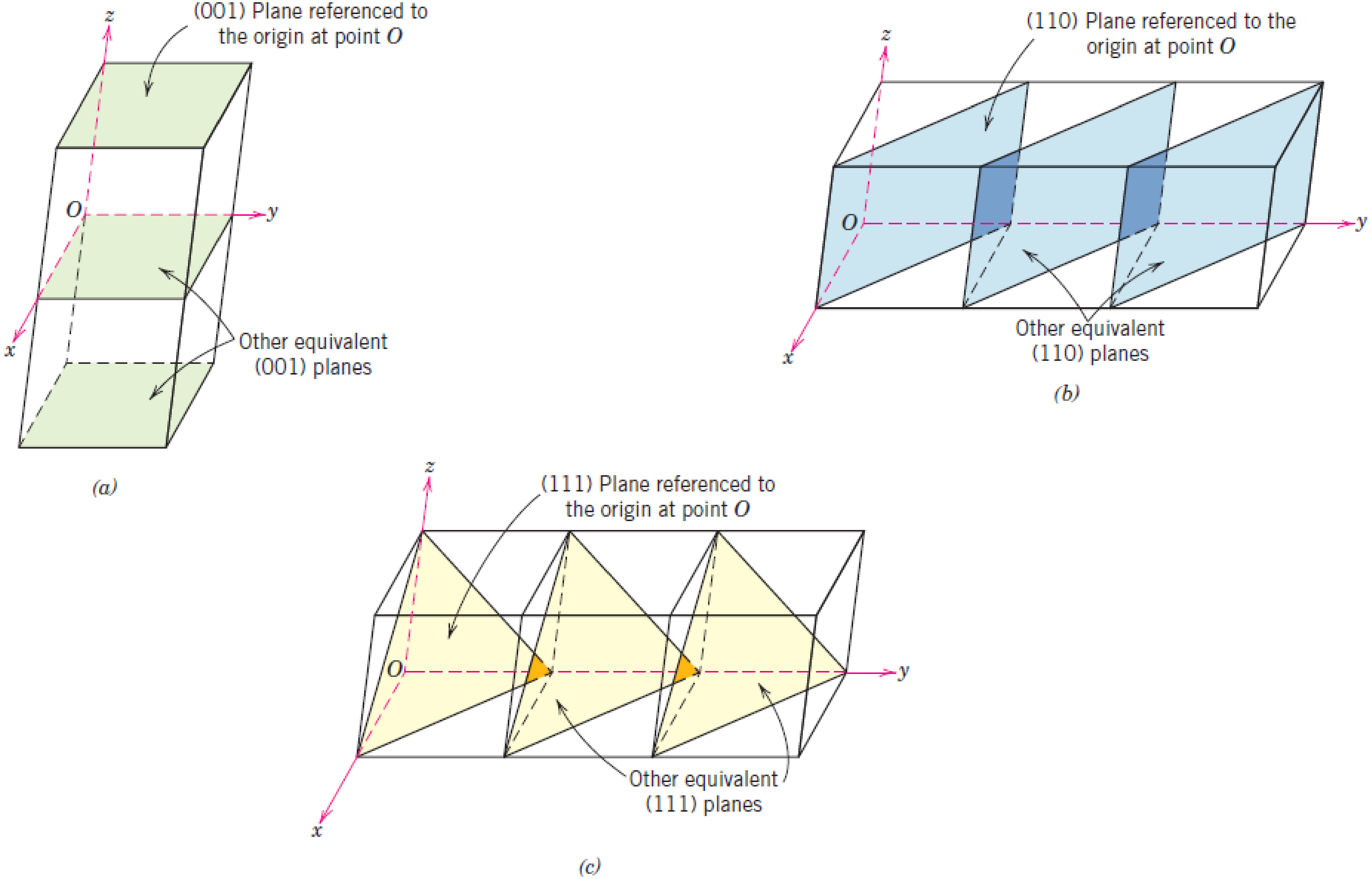


Crystallographic Planes

The procedure used to determine the h, k, and l index numbers is as follows:

1. If the plane passes through the selected origin, either another parallel plane must be constructed within the unit cell by an appropriate translation, or a new origin must be established at the corner of another unit cell.
2. At this point the crystallographic plane either intersects or parallels each of the three axes; the length of the planar intercept for each axis is determined in terms of the lattice parameters a, b, and c.
3. The reciprocals of these numbers are taken. A plane that parallels an axis may be considered to have an infinite intercept, and, therefore, a zero index.
4. If necessary, these three numbers are changed to the set of smallest integers by multiplication or division by a common factor.3
5. Finally, the integer indices, not separated by commas, are enclosed within parentheses, thus: (hkl).

Crystallographic Planes

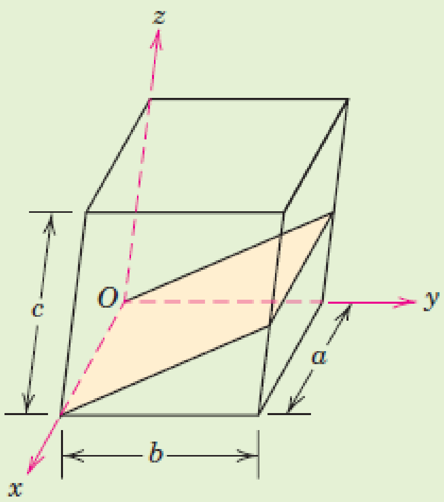


Crystallographic Planes

EXAMPLE PROBLEM 3.10

Determination of Planar (Miller) Indices

Determine the Miller indices for the plane shown in the accompanying sketch (*a*).



(*a*)

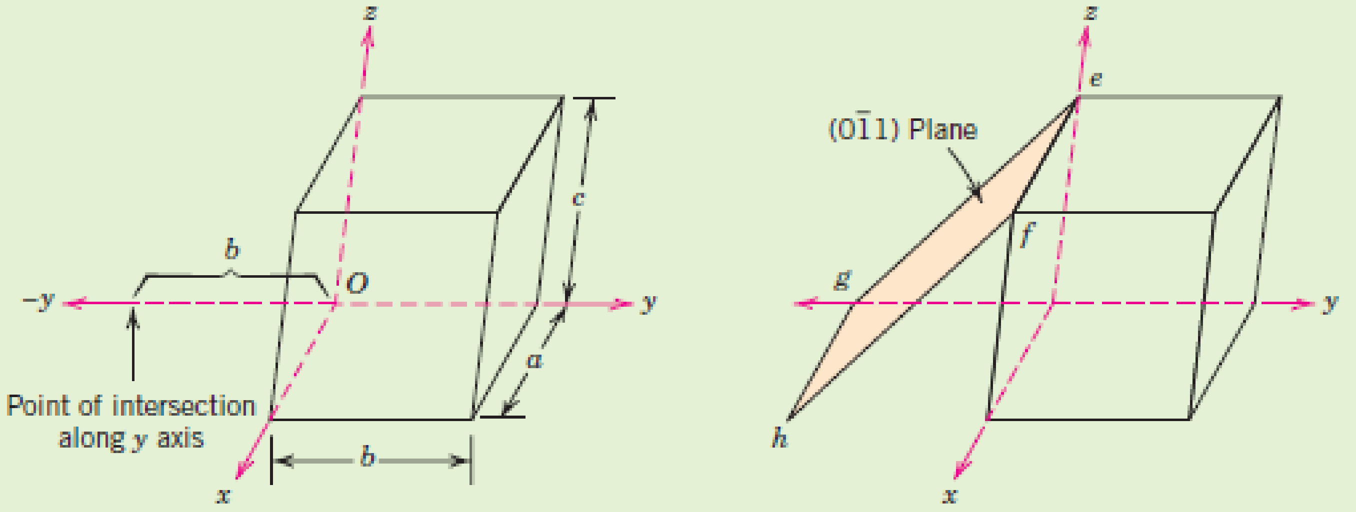
	x	y	z
Intercepts	∞a	$-b$	$c/2$
Intercepts (in terms of lattice parameters)	∞	-1	$\frac{1}{2}$
Reciprocals	0	-1	2
Reductions (unnecessary)			
Enclosure		$(0\bar{1}2)$	

Crystallographic Planes

EXAMPLE PROBLEM 3.11

Construction of Specified Crystallographic Plane

Construct a $(0\bar{1}1)$ plane within a cubic unit cell.



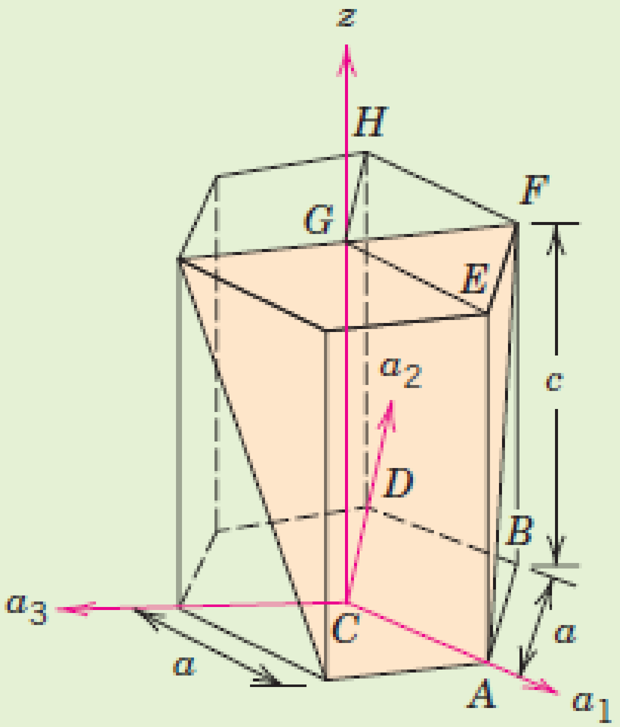
(a) (b)

Crystallographic Planes

EXAMPLE PROBLEM 3.12

Determination of Miller–Bravais Indices for a Plane within a Hexagonal Unit Cell

Determine the Miller–Bravais indices for the plane shown in the hexagonal unit cell.



Solution

To determine these Miller–Bravais indices, consider the plane in the figure referenced to the parallelepiped labeled with the letters *A* through *H* at its corners. This plane intersects the a_1 axis at a distance *a* from the origin of the a_1-a_2-a_3-z coordinate axis system (point *C*). Furthermore, its intersections with the a_2 and *z* axes are $-a$ and *c*, respectively. Therefore, in terms of the lattice parameters, these intersections are 1, −1, and 1. Furthermore, the reciprocals of these numbers are also 1, −1, and 1. Hence

$$h = 1$$
$$k = -1$$
$$l = 1$$

and, from Equation 3.7,

$$\begin{aligned} i &= -(h + k) \\ &= -(1 - 1) = 0 \end{aligned}$$

Therefore the $(hkil)$ indices are $(1\bar{1}01)$.

Notice that the third index is zero (i.e., its reciprocal = ∞), which means that this plane parallels the a_3 axis. Inspection of the preceding figure shows that this is indeed the case.

Linear Densities

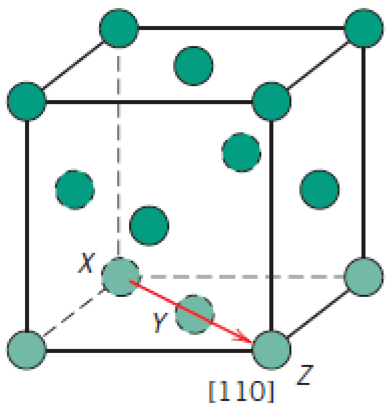
Linear density (LD) is defined as the number of atoms per unit length whose centers lie on the direction vector for a specific crystallographic direction;

$$\mathrm{LD} = \frac{\text{number of atoms centered on direction vector}}{\text{length of direction vector}}$$

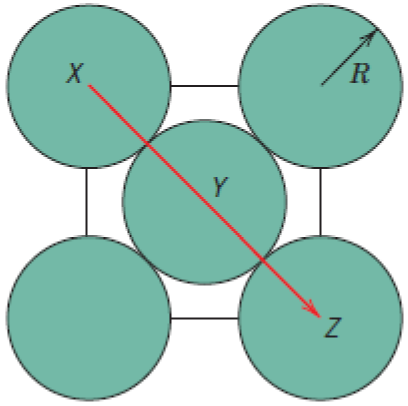
Determine the linear density of the [110] direction for the FCC crystal structure

<110> directions in the FCC lattice have 2 atoms (1/2 x 2 corner atoms + 1 center atom) and the length is 4R

$$LD_{110} = \frac{2 atoms}{4R} = \frac{1}{2R} = \frac{\sqrt{2}}{a}$$



(a)



(b)

$a = 2R\sqrt{2}$

Planar Densities

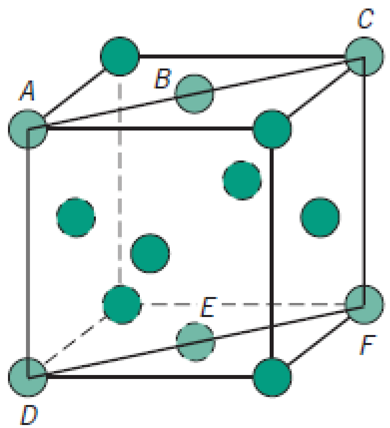
Planar density (PD) is taken as the number of atoms per unit area that are centered on a particular crystallographic plane

$$PD = \frac{\text{number of atoms centered on a plane}}{\text{area of plane}}$$

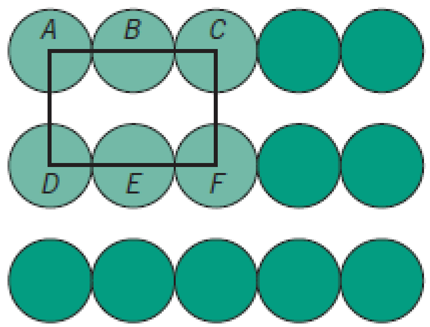
For eg. Planar density of {110} planes in the FCC crystal

There are 2 atoms (1/4 x 4 corner atoms + 1/2 x 2 side atoms) in the {110} planes in the FCC lattice.

$$PD_{110} = \frac{2atoms}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}} = \frac{\sqrt{2}}{a^2}$$



(a)



(b)

$a = 2R\sqrt{2}$

Crystalline and Noncrystalline Materials

Single crystals

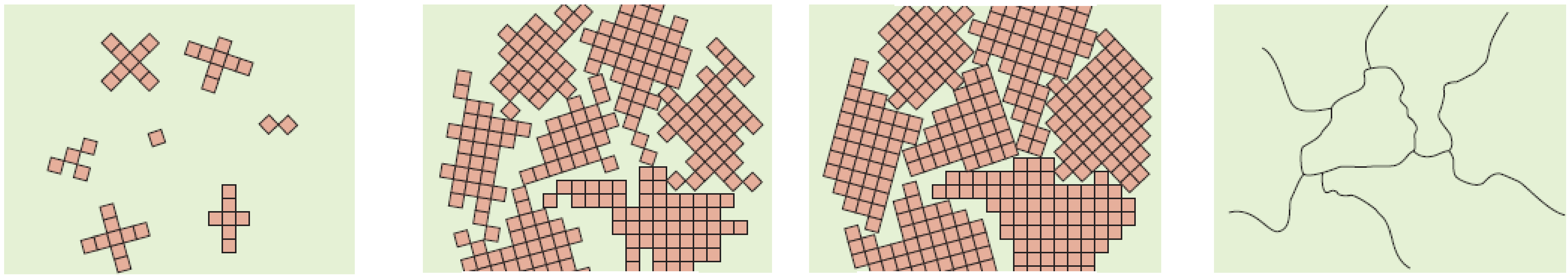
- For a crystalline solid, when the periodic and repeated arrangement of atoms is perfect or extends throughout the entirety of the specimen without interruption, the result is a ***single crystal***.
- All unit cells interlock in the same way and have the same orientation.
- single crystals have become extremely important in many of our modern technologies, in particular *electronic microcircuits*, which employ single crystals of silicon and other semiconductors

Crystalline and Noncrystalline Materials

Polycrystalline materials

- Most crystalline solids are composed of a collection of many small crystals or grains; such materials are termed ***polycrystalline***.

<u>Schematic diagrams of the various stages in the solidification of a polycrystalline material;</u>



(a) Small crystallite nuclei.

(b) Growth of the crystallites; the obstruction of some grains that are adjacent to one another.

(c) Upon completion of solidification, grains having irregular shapes have formed.

(d) The grain structure as it would appear under the microscope; dark lines are the ***grain boundaries***.

Crystalline and Noncrystalline Materials

- Anisotropy

The physical properties of single crystals of some substances depend on the crystallographic direction in which measurements are taken. This directionality of properties is termed ***anisotropy***, and it is associated with the variance of atomic or ionic spacing with crystallographic direction

Eg., the elastic modulus, the electrical conductivity, and the index of refraction may have different values in the [100] and [111] directions.

	Modulus of Elasticity (GPa)		
Metal	[*100*]	[*110*]	[*111*]
Aluminum	63.7	72.6	76.1
Copper	66.7	130.3	191.1
Iron	125.0	210.5	272.7
Tungsten	384.6	384.6	384.6

- Substances in which measured properties are independent of the direction of measurement are ***isotropic***.

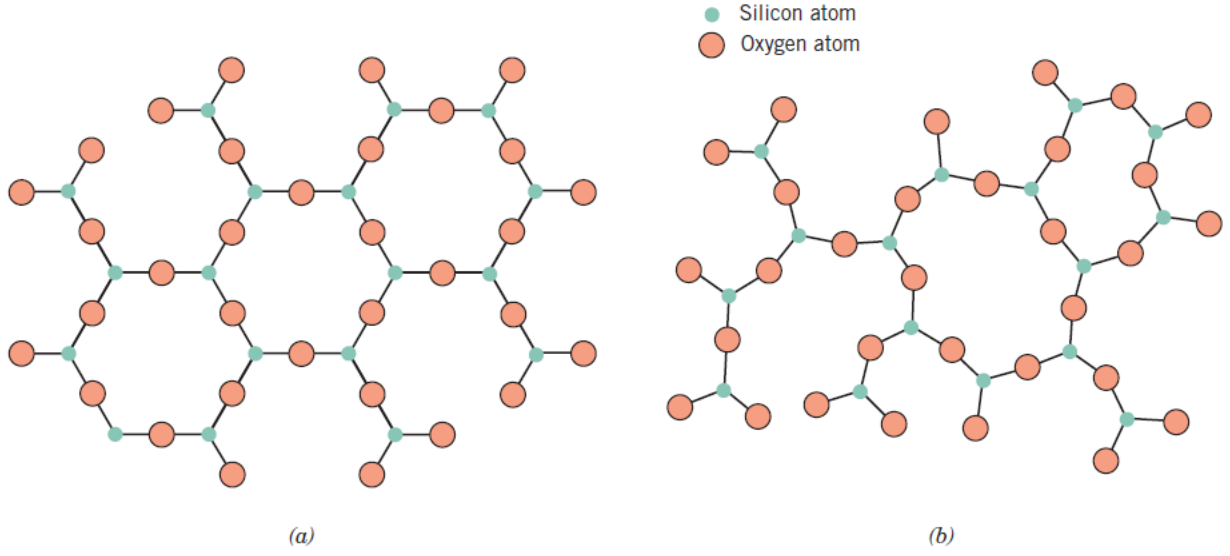
Crystalline and Noncrystalline Materials

Noncrystalline Solids

lack a systematic and regular arrangement of atoms over relatively large atomic distances. Sometimes such materials are also called ***amorphous***

<u>Two-dimensional schemes of the structure of *silicon dioxide*</u>

each silicon ion bonds to three oxygen ions for both states, beyond this, the structure is much more disordered and irregular for the non-crystalline structure.



<u>crystalline</u>

<u>Noncrystalline</u>

Summary

Fundamental Concepts: Atoms in crystalline solids are positioned in orderly and repeated patterns that are in contrast to the random and disordered atomic distribution found in noncrystalline or amorphous materials.

Unit Cells: Crystal structures are specified in terms of parallelepiped unit cells, which are characterized by geometry and atom positions within.

Metallic Crystal Structures: Most common metals exist in at least one of three relatively simple crystal structures:

- Face-centered cubic (FCC), which has a cubic unit cell.
- Body-centered cubic (BCC), which also has a cubic unit cell.
- Hexagonal close-packed, which has a unit cell of hexagonal symmetry.

- Unit cell edge length (a) and atomic radius (R) are related (See Equations for face-centered cubic, and body-centered cubic.
- Two features of a crystal structure are
  - Coordination number—the number of nearest-neighbor atoms, and
  - Atomic packing factor—the fraction of solid sphere volume in the unit cell.

Summary

Density Computations: The theoretical density of a metal (ρ) is a function of the number of equivalent atoms per unit cell, the atomic weight, unit cell volume, and Avogadro's number.

Polymorphism and Allotropy

Polymorphism is when a specific material can have more than one crystal structure.

Allotropy is polymorphism for elemental solids.

Crystal Systems

The concept of crystal system is used to classify crystal structures on the basis of unit cell geometry—that is, unit cell edge lengths and interaxial angles. There are seven crystal systems: cubic, tetragonal, hexagonal, orthorhombic, rhombohedral, (trigonal), monoclinic, and triclinic.

Summary

Point Coordinates, Crystallographic Directions, Crystallographic Planes

Crystallographic points, directions, and planes are specified in terms of indexing schemes. The basis for the determination of each index is a coordinate axis system defined by the unit cell for the particular crystal structure.

- The location of a point within a unit cell is specified using coordinates that are fractional multiples of the cell edge lengths.
- Directional indices are computed in terms of the vector projection on each of the coordinate axes.
- Planar (or Miller) indices are determined from the reciprocals of axial intercepts.

For hexagonal unit cells, a four-index scheme for both directions and planes is found to be more convenient.

Summary

Single Crystals, Polycrystalline Materials

Single crystals are materials in which the atomic order extends uninterrupted over the entirety of the specimen; under some circumstances, single crystals may have flat faces and regular geometric shapes.

The vast majority of crystalline solids, however, are polycrystalline, being composed of many small crystals or grains having different crystallographic orientations.

A grain boundary is the boundary region separating two grains, wherein there is some atomic mismatch.

Anisotropy

Anisotropy is the directionality dependence of properties. For isotropic materials, properties are independent of the direction of measurement.